

DM42

NUMERICAL INTEGRATION HANDBOOK

From Simple Areas to Advanced Scientific Modelling

A practical, evidence-based guide to $\int f(x)$, ACC, numerical strategy,
and defensible scientific use on the SwissMicros DM42

Research snapshot: 2 October 2026

How to use this handbook

Begin with Parts I–IV if $f(x)$ is new. Experienced users can start with the ACC chapter, the interval-splitting chapters, or the advanced case studies. Every worked result has an independent high-precision reference; representative X/Y pairs are modeled from the actual Free42 3.3.10 integration loop used by published DM42 firmware v3.26.

KEY IDEA Never merely accept the integral. Predict its sign, scale and units; inspect Y; change ACC; split the interval; and compare against an independent fact whenever the answer matters.

Scope and versioning. SwissMicros' DM42 User Manual is v3.42 dated 24 September 2026. The latest published DM42 program firmware found for this research snapshot is v3.26 (14 November 2025), built with Free42 core 3.3.10. The handbook separates manual version from firmware version deliberately. Use IDENT on your calculator to identify the installed core when reproducing edge cases.

Numerical results. The reference values were recalculated with 60-decimal-digit arithmetic. The reported Y values are faithful algorithm-level predictions, not captured hardware screenshots; final decimal digits and running time can vary with firmware, mode and the transcendental library. The physical interpretation is only as good as the model and input data.

Keystroke notation

Notation	Meaning	Example
[KEY]	physical key or named shifted function	[PRGM], [RCL], [$f(x)$]
{SOFT}	top-row soft key carrying the displayed menu label	{LLIM}, {ACC}, {f}
⟨TEXT⟩	type the literal alpha text, then ENTER	⟨MVAR⟩, ⟨GAUSS⟩
→	perform the next action; not a calculator instruction	0 {LLIM} → 1 {ULIM}
display: ...	text or value expected on the calculator	display: $f=0.5$
01 ...	program line exactly as listed	01 LBL "LIN"

A bracketed shifted legend such as [$f(x)$] means press SHIFT, then the physical key carrying that legend. Menu labels are shown in braces because their physical top-row key depends on the label currently displayed. In Program-entry mode, enter ordinary instructions with their physical key or menu. Pressing [XEQ] begins an actual XEQ instruction; it is used only when a program must call a labelled subroutine.

The working discipline

1. Name the quantity being accumulated.
2. Choose the integration variable and differential.
3. Write the integrand and state its units.
4. Choose finite limits and inspect the function for poles, jumps, peaks and cancellation.
5. Estimate sign and order of magnitude.
6. Choose ACC to match the data and purpose.
7. Run $\int f(x)$, read X and inspect Y.
8. Repeat with tighter ACC and, when appropriate, split intervals.
9. Verify independently and report only meaningful digits.

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PART I

WHAT INTEGRATION MEANS

From tiny contributions to signed area, accumulated physics and dimensional sense.

1 Accumulation before symbols

Integration adds contributions that vary continuously. If a rate $q(x)$ applies over a tiny interval dx , the tiny contribution is approximately $q(x)dx$. Adding many such pieces and taking the limiting process produces a definite integral. The differential is not decoration: it names the variable whose tiny intervals are being accumulated.

$$\text{quantity accumulated} = \int_a^b \text{rate-per-unit-}x \times dx$$

Situation	Tiny contribution	Integral	Resulting unit
motion	$v(t) dt$	$s = \int v(t)dt$	$(m/s) \cdot s = m$
work	$F(x) dx$	$W = \int F(x)dx$	$N \cdot m = J$
charge	$I(t) dt$	$Q = \int I(t)dt$	$A \cdot s = C$
mass	ρdV	$m = \int \rho dV$	$(kg/m^3) \cdot m^3 = kg$
probability	$f(x) dx$	$P = \int f(x)dx$	dimensionless

UNITS CHECK If integrating force with respect to time, the answer is impulse, not work. Work requires displacement as the integration variable. The variable and differential determine the physical meaning.

2 Definite, indefinite and numerical integrals

Idea	What it asks for	Typical output
indefinite integral	a family of antiderivatives	$F(x)+C$
definite integral	one accumulated value between limits	a number with units
symbolic integration	an exact algebraic antiderivative when available	expression
numerical quadrature	an approximation from sampled $f(x)$ values	number + estimated numerical uncertainty

The DM42 $\int f(x)$ application performs numerical quadrature. It repeatedly calls a small program that returns $f(x)$. It does not ask that program to know, construct or store an antiderivative.

$$\text{Fundamental Theorem: if } F'(x)=f(x), \text{ then } \int_a^b f(x)dx = F(b)-F(a)$$

The theorem explains why antiderivatives can evaluate definite integrals. Numerical integration follows another route: it approximates the accumulation directly from function values.

3 Signed area is not geometric area

Contributions above the x-axis are positive; those below are negative. Therefore a zero integral can hide equal nonzero lobes.

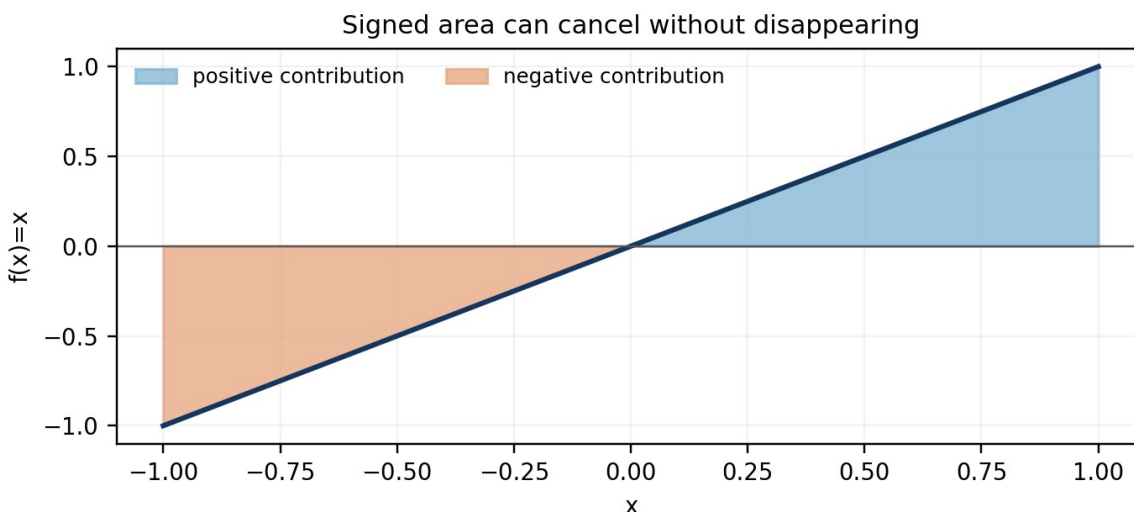


Figure 3.1 $\int_{-1}^1 x \, dx = 0$, but the geometric area is $\int_{-1}^1 |x| \, dx = 1$.

$$\int_{-1}^1 x \, dx = 0 \quad \text{while} \quad \int_{-1}^1 |x| \, dx = 1$$

KEY IDEA Always decide whether the problem asks for signed accumulation, total geometric area, absolute exposure, or some other measure. They are different mathematical quantities.

PART II

HOW NUMERICAL INTEGRATION WORKS

Build intuition with panels, then look at the verified current DM42 algorithm.

4 From rectangles to refinement

Rectangles estimate each narrow strip from one height. Trapezoids join neighboring samples by straight lines. Simpson-type rules use local curvature information. Refinement repeats the estimate with smaller spacing; a stable sequence of estimates is evidence—never proof—that the quadrature error is shrinking.

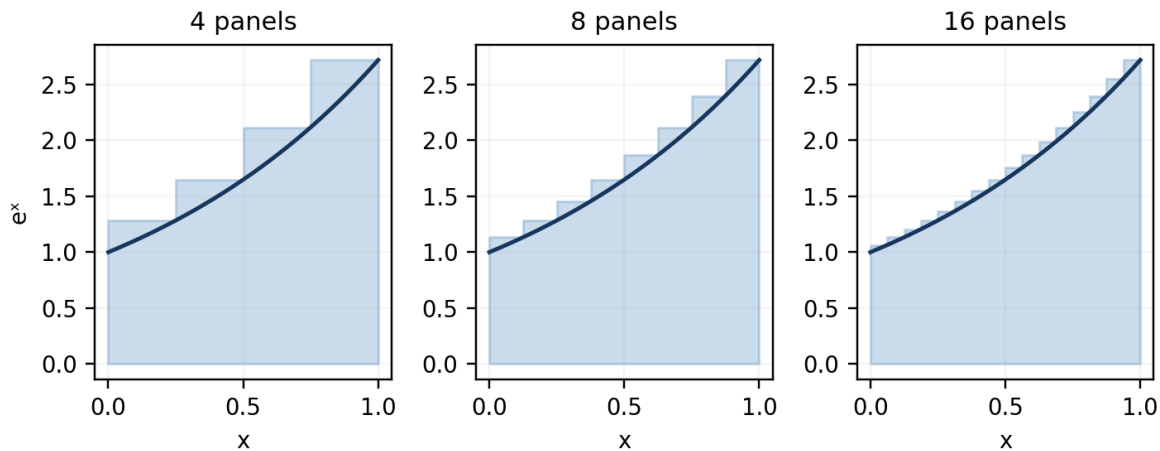


Figure 4.1 More panels reveal more of the function. These rectangles teach intuition; they are not a claim about the DM42 implementation.

Method idea	Uses	Strength	Weakness
rectangles	one height per panel	simple intuition	slow accuracy improvement
trapezoids	straight chord per panel	reuses refined samples	curvature remains
Simpson-type	local quadratic behavior	high order on smooth functions	not the verified current DM42 method
Romberg	trapezoid-like refinement + extrapolation	rapid on smooth functions	can be fooled by unresolved features

5 What current published DM42 firmware does

SwissMicros firmware history identifies DM42 v3.26 as using Free42 core 3.3.10. The matching published source implements INTEG with Romberg extrapolation and the substitution $x=(3u-u^3)/2$ mapped to the requested interval. The source comment states that this prevents endpoint evaluation and creates nonuniform sampling.

$$u \mapsto (3u-u^3)/2; \text{ then map } [-1,1] \text{ to } [LLIM,ULIM]$$

10. Evaluate the transformed weighted integrand at a first set of interior points.
11. Double the sample count at each refinement level.
12. Retain five successive approximations and apply Romberg/Richardson extrapolation.
13. Form a new extrapolated result and compare it with the previous extrapolated result.
14. Stop when the difference is no greater than $ACC \times |result|$, or when the implementation reaches its refinement cap.

The core defines a maximum corresponding to 524,287 function evaluations. That cap prevents infinite work; it does not make a difficult integral valid. The endpoint-avoiding substitution is helpful for some endpoint behavior, but it does not magically regularize a pole or guarantee that a narrow peak was sampled.

ACCURACY MATTERS γ is set to the absolute difference between the last two extrapolated estimates in the current source. Treat it as a convergence indicator for this algorithm, not a universal bound on the true mathematical error.

6 Romberg integration without the fog

Suppose a sequence of refined approximations is $T_0, T_1, T_2, \dots$. For a sufficiently smooth integrand, much of the leading error behaves predictably as spacing is halved. Richardson extrapolation combines neighboring estimates to cancel leading error terms. Romberg integration repeats this idea in a triangular table. The DM42 current core uses a rolling five-estimate window rather than displaying that table.

$$R_{k,0} = T_k \quad \text{and} \quad R_{k,j} = R_{k,j-1} + (R_{k,j-1} - R_{k-1,j-1}) / (4^j - 1)$$

This formula is explanatory. The calculator's transformed rule and rolling implementation are the authority for its actual numbers. Smoothness matters because extrapolation assumes an orderly error pattern.

PART III

THE DM42 $\int f(x)$ APPLICATION

Define the integrand, choose the variable, control ACC and interpret the X/Y result.

7 The complete interactive workflow

15. Create a new program space with [GTO] [.] [.] and enter Program-entry mode with [PRGM].
16. Insert a global LBL, contiguous MVAR declarations, and instructions that leave $f(x)$ in X. The existing .END. terminates the program.
17. Leave Program-entry mode with [PRGM].
18. Press [$\int f(x)$] and select the integrand program.
19. Store each fixed parameter by typing its value and pressing its soft key.
20. Select the integration variable by pressing its soft key with no number pending.
21. Store LLIM, ULIM and ACC.
22. Press [$\int$]. Read the integral in X and use [$X \leftrightarrow Y$] to inspect the returned convergence estimate in Y.

DM42 TECHNIQUE Typing a number then pressing an MVAR soft key stores the number. Pressing that soft key with no number pending selects the variable of integration. These are intentionally different actions.

8 The minimum integrand program

```
01 LBL "MODEL"
02 MVAR "X"
03 MVAR "A"
04 MVAR "B"
05 ... calculate f(X,A,B) ...
06 END
```

- LBL makes the program selectable from the $\int f(x)$ application.
- MVAR builds the variable menu. The selected integration variable is changed internally at every sample; the others remain parameters.
- The body is ordinary RPN evaluation. It must finish with one real value $f(x)$ in X.
- Use short global labels and variable names (up to seven characters in this calculator family).

9 Stack state after integration

Register	Verified current meaning	Advice
X	integral approximation	the primary result
Y	last-change / convergence estimate returned by INTEG	inspect with $X \leftrightarrow Y$; do not call it a guarantee
Z, T	affected by ordinary two-result stack-lift behavior	do not build a workflow that assumes old contents survive

The current core creates two results and places approximation in X and estimate in Y. In four-level stack mode, the older stack is shifted according to the ordinary stack-lift state; in big-stack mode two results are pushed unless stack-lift was disabled. The portable promise from current documentation is therefore X and Y, not a fixed story for Z and T.

10 ACC: requested convergence, not truth

Historically, HP described ACC as an estimate of the relative error in each computed $f(x)$, connecting it conceptually to an uncertainty proportional to $\text{ACC} \int |f(x)| dx$. That remains valuable scientific advice: do not request precision finer than the integrand's data and model. The current Free42 3.3.10 implementation used by DM42 v3.26 has a directly inspectable stopping test:

$$Y = |I_n - I_{n-1}| \quad \text{and stop when} \quad Y \leq \text{ACC} \cdot |I_n|$$

Thus, for current firmware, ACC behaves operationally as a relative convergence threshold. A small result produced by cancellation can make this criterion demanding; a large result can make the allowed absolute change larger. Actual error can still exceed Y when the estimates converge to the wrong value or miss a feature.

ACC	Modeled X for $\int_0^\pi \sin x \, dx$	Modeled Y	f evaluations	$ X - 2 $
1e-4	2.000000322051277	9.943313e-5	31	3.2205e-7
1e-8	2.0000000000000122	1.126684e-10	127	1.2204e-13
1e-12	2.0000000000000000	1.219234e-13	255	1.1701e-16
1e-16	2.0000000000000000	1.168920e-16	511	1.1300e-19

The pattern is more important than the final digits: tighter ACC usually increases work and stabilizes X. But Y and the true error are different quantities. In the first row, for example, the reported last change and the actual error are of the same order but not equal.

10.1 What ACC=0 means now

Current core source does not interpret zero as an abstract request for exact arithmetic. It clamps ACC below an internal floor. In the decimal build used by the DM42, that floor is approximately $1\text{E}-33$ (ten times the spacing immediately below 1 in decimal128). ACC above 1 is clamped to 1. A missing ACC is treated like zero and then raised to the same floor.

NUMERICAL TRAP ACC=0 can force many refinements without creating more trustworthy physics. Use it only as a deliberate convergence experiment on a well-behaved, well-scaled integrand. Save calculator state first when experimenting with costly problems.

Situation	Sensible ACC strategy
classroom analytic check	start around $1\text{E}-8$; tighten once and compare
measured laboratory data	choose ACC comfortably below measurement/model uncertainty, not dozens of orders below it
engineering estimate	often $1\text{E}-5$ to $1\text{E}-8$, followed by a split-interval check
high-precision smooth mathematics	progressively tighten; compare X, Y and an independent value

Situation	Sensible ACC strategy
poorly known empirical model	let input/model uncertainty determine meaningful digits; a tiny ACC adds no knowledge
near-zero cancellation	integrate lobes separately or exploit symmetry; examine absolute errors

11 Reading an answer responsibly

Layer	Question
mathematical	Does the integral exist and match the quantity intended?
quadrature	Did X stabilize under tighter ACC and useful interval splits?
calculator	Is Y small enough relative to the reporting goal?
data	Are constants and measurements known to this precision?
model	Does the simplified model represent the real system well enough?

ACCURACY MATTERS A display such as 3.728194572814 does not justify twelve decimals. If the governing coefficient is known to roughly 0.5%, the scientifically honest report may be 3.73, even when the numerical integrator is far more repeatable.

PART IV

YOUR FIRST INTEGRALS

Deliberately easy mathematics so every key and every register makes sense.

12 We know the answer—so test the integrator

The next problems are intentionally simple. Their known answers isolate calculator technique from mathematical uncertainty. Follow every entry step for the first problem; then repeat until the $\int f(x)$ rhythm becomes automatic.

12.1 The first integral: area under $f(x)=x$

Accumulate the height x from 0 to 1. The triangle gives an answer known before the calculator is touched.

$$I = \int_0^1 x \, dx = 1/2$$

Variable	Meaning	Unit
X	integration variable	1

Item	Value
Integration variable	X
LLIM	0
ULIM	1
ACC	1E-10
Expected scale	about 0.5

Integrand program

```
01 LBL "LIN"
02 MVAR "X"
03 RCL "X"
04 END
```

Program-entry keys

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

Line 01: [PGM.FCN] {LBL} {LIN} [ENTER]

Line 02: [PGM.FCN] [▲] {MVAR} {X} [ENTER]

Line 03: [RCL] {X} [ENTER]

The new space already ends in .END.; finish: [PRGM]

Run $\int f(x)$

[$\int f(x)$] {LIN} → {X} → [0] {LLIM} → [1] {ULIM} → [1] [E] [+/-] [1] [0] {ACC} → {f}

Read X for the integral. Press [X<>Y] to inspect Y, then [X<>Y] again to restore X.

Register / check	Value
X — integral	0.5

Register / check	Value
Y — returned convergence estimate	approximately 0
Independent reference	0.5; $ \Delta \approx 0$

VERIFICATION 0.5; $|\Delta| \approx 0$. The modeled Y value is an algorithmic estimate from the current Free42 3.3.10 loop, not a proof that the model or data are correct.

KEY IDEA The integrator needs numerical values of $f(x)$, not an antiderivative.

12.2 A constant integrand

The integral of 1 over a width of 2 must equal 2.

$$I = \int_0^2 1 \, dx = 2$$

Variable	Meaning	Unit
X	integration variable (unused by the constant)	1

Item	Value
Integration variable	X
LLIM	0
ULIM	2
ACC	1E-10
Expected scale	exactly 2

Integrand program

```
01 LBL "ONE"
02 MVAR "X"
03 1
04 END
```

Program-entry keys

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

Line 01: [PGM.FCN] {LBL} {ONE} [ENTER]

Line 02: [PGM.FCN] [▲] {MVAR} {X} [ENTER]

Line 03: [1]

The new space already ends in .END.; finish: [PRGM]

Run $\int f(x)$

[f(x)] {ONE} → {X} → [0] {LLIM} → [2] {ULIM} → [1] [E] [+/-] [1] [0] {ACC} → {f}

Read X for the integral. Press [X<>Y] to inspect Y, then [X<>Y] again to restore X.

Register / check	Value
X — integral	2
Y — returned convergence estimate	approximately 0
Independent reference	2; $ \Delta \approx 0$

VERIFICATION 2; $|\Delta| \approx 0$. The modeled Y value is an algorithmic estimate from the current Free42 3.3.10 loop, not a proof that the model or data are correct.

KEY IDEA A declared integration variable may be absent from a constant integrand.

12.3 Same integrand, different limits

Reuse LIN from 0 to 3.

$$I = \int_0^3 x \, dx = 9/2$$

Variable	Meaning	Unit
X	integration variable	1

Item	Value
Integration variable	X
LLIM	0
ULIM	3
ACC	1E-10
Expected scale	4.5

Integrand program

```
01 LBL "LIN"
02 MVAR "X"
03 RCL "X"
04 END
```

Program-entry keys

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

Line 01: [PGM.FCN] {LBL} {LIN} [ENTER]

Line 02: [PGM.FCN] [▲] {MVAR} {X} [ENTER]

Line 03: [RCL] {X} [ENTER]

The new space already ends in .END.; finish: [PRGM]

Run $\int f(x)$

`[ $\int f(x)$ ] {LIN}  $\rightarrow$  {X}  $\rightarrow$  [0] {LLIM}  $\rightarrow$  [3] {ULIM}  $\rightarrow$  [1] [E] [+/-] [1] [0] {ACC}  $\rightarrow$  {f}`
 Read X for the integral. Press `[X<>Y]` to inspect Y, then `[X<>Y]` again to restore X.

Register / check	Value
X — integral	4.5
Y — returned convergence estimate	approximately 0
Independent reference	4.5; $ \Delta \approx 0$

VERIFICATION 4.5; $|\Delta| \approx 0$. The modeled Y value is an algorithmic estimate from the current Free42 3.3.10 loop, not a proof that the model or data are correct.

KEY IDEA Limits belong to the integration request, so the same program is reusable.

12.4 A quadratic

Integrate x^2 from 0 to 2 and compare with $x^3/3$.

$$I = \int_0^2 x^2 dx = 8/3$$

Variable	Meaning	Unit
X	integration variable	1

Item	Value
Integration variable	X
LLIM	0
ULIM	2
ACC	1E-10
Expected scale	about 2.67

Integrand program

```
01 LBL "SQR"
02 MVAR "X"
03 RCL "X"
04 X $\uparrow$ 2
05 END
```

Program-entry keys

Start: `[GT0] [.] [.] [PRGM]` (new program space, then Program-entry mode)

Line 01: `[PGM.FCN] {LBL} {SQR} [ENTER]`

Line 02: `[PGM.FCN] [ $\blacktriangle$ ] {MVAR} {X} [ENTER]`

Line 03: [RCL] (X) [ENTER]

Line 04: [X+2]

The new space already ends in .END.; finish: [PRGM]

Run $\int f(x)$

[$\int f(x)$] {SOR} → {X} → [0] {LLIM} → [2] {ULIM} → [1] [E] [+/-] [1] [0] {ACC} → {f}

Read X for the integral. Press [X<>Y] to inspect Y, then [X<>Y] again to restore X.

Register / check	Value
X — integral	2.6666666667
Y — returned convergence estimate	approximately 0
Independent reference	2.6666666667; $ \Delta \approx 0$

VERIFICATION 2.6666666667; $|\Delta| \approx 0$. The modeled Y value is an algorithmic estimate from the current Free42 3.3.10 loop, not a proof that the model or data are correct.

KEY IDEA Polynomial checks expose entry errors immediately.

12.5 Sine over a half-cycle

In radians, positive sine area from 0 to π is 2.

$$I = \int_0^{\pi} \sin x \, dx = 2$$

Variable	Meaning	Unit
X	angle and integration variable	rad

Item	Value
Integration variable	X
LLIM	0
ULIM	π
ACC	1E-10
Expected scale	2

Integrand program

```

01 LBL "SINE"
02 MVAR "X"
03 RCL "X"
04 SIN
05 END

```

Program-entry keys

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

Line 01: [PGM.FCN] {LBL} {SINE} [ENTER]

Line 02: [PGM.FCN] [▲] {MVAR} {X} [ENTER]

Line 03: [RCL] {X} [ENTER]

Line 04: [SIN]

The new space already ends in .END.; finish: [PRGM]

Run $\int f(x)$

[MODES] {RAD} → [∫f(x)] {SINE} → {X} → [0] {LLIM} → [π] {ULIM} → [1] [E] [+/-] [1] [0] {ACC} → {f}

Read X for the integral. Press [X<>Y] to inspect Y, then [X<>Y] again to restore X.

Register / check	Value
X — integral	2
Y — returned convergence estimate	approximately 1.13e-10
Independent reference	2; $ \Delta \approx 1.22\text{e-}13$

VERIFICATION 2; $|\Delta| \approx 1.22\text{e-}13$. The modeled Y value is an algorithmic estimate from the current Free42 3.3.10 loop, not a proof that the model or data are correct.

What can go wrong

- DEG mode evaluates a different function.

KEY IDEA Angle mode is part of the mathematical model, not a display preference.

12.6 Cosine over a quarter-cycle

In radians, integrate cosine from 0 to $\pi/2$.

$$I = \int_0^{\pi/2} \cos x \, dx = 1$$

Variable	Meaning	Unit
X	angle and integration variable	rad

Item	Value
Integration variable	X
LLIM	0
ULIM	$\pi \div 2$
ACC	1E-10
Expected scale	1

Integrand program

```

01 LBL "COSIN"
02 MVAR "X"
03 RCL "X"
04 COS
05 END

```

Program-entry keys

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

Line 01: [PGM.FCN] {LBL} {COSIN} [ENTER]

Line 02: [PGM.FCN] [▲] {MVAR} {X} [ENTER]

Line 03: [RCL] {X} [ENTER]

Line 04: [COS]

The new space already ends in .END.; finish: [PRGM]

Run $\int f(x)$

[MODES] {RAD} → [∫f(x)] {COSIN} → {X} → [0] {LLIM} → [π] [ENTER] [2] [÷] {ULIM} → [1] [E]
 [+/-] [1] [0] {ACC} → {f}

Read X for the integral. Press [X<>Y] to inspect Y, then [X<>Y] again to restore X.

Register / check	Value
X — integral	1
Y — returned convergence estimate	approximately 3.64e-12
Independent reference	1; $ \Delta \approx 2.8\text{e-}15$

VERIFICATION 1; $|\Delta| \approx 2.8\text{e-}15$. The modeled Y value is an algorithmic estimate from the current Free42 3.3.10 loop, not a proof that the model or data are correct.

What can go wrong

- Do not type 90 unless the program and limits are intentionally in degrees.

KEY IDEA Use the stack to calculate $\pi/2$ before storing ULIM.

12.7 Logarithmic area without an antiderivative program

Integrate $1/x$ from 1 to e.

$$I = \int_1^e (1/x) \, dx = 1$$

Variable	Meaning	Unit
X	positive integration variable	1

Item	Value
Integration variable	X

Item	Value
LLIM	1
ULIM	e
ACC	1E-10
Expected scale	1

Integrand program

```
01 LBL "RECIP"
02 MVAR "X"
03 RCL "X"
04 1/X
05 END
```

Program-entry keys

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

Line 01: [PGM.FCN] {LBL} {RECIP} [ENTER]

Line 02: [PGM.FCN] [▲] {MVAR} {X} [ENTER]

Line 03: [RCL] {X} [ENTER]

Line 04: [1/X]

The new space already ends in .END.; finish: [PRGM]

Run $\int f(x)$

[$\int f(x)$] {RECIP} → {X} → [1] {LLIM} → [1] [E+X] {ULIM} → [1] [E] [+/-] [1] [0] {ACC} → {f}

Read X for the integral. Press [X<>Y] to inspect Y, then [X<>Y] again to restore X.

Register / check	Value
X — integral	1
Y — returned convergence estimate	approximately 8.98e-15
Independent reference	1; $ \Delta \approx 1.75e-17$

VERIFICATION 1; $|\Delta| \approx 1.75e-17$. The modeled Y value is an algorithmic estimate from the current Free42 3.3.10 loop, not a proof that the model or data are correct.

What can go wrong

- Crossing $x=0$ changes the problem into a divergent improper integral.

KEY IDEA The interval must stay inside the function's domain.

PART V

TRIVIAL APPLIED EXAMPLES

Build fluency with units, parameters, limits, ACC and X/Y interpretation.

13 Accumulation in everyday models

13.1 Distance at constant velocity

A vehicle travels at 12 m/s for 8 s.

$$s = \int_0^8 v \, dt$$

Variable	Meaning	Unit
T	time	s
V	constant velocity	m/s

Item	Value
Integration variable	T
LLIM	0
ULIM	8
ACC	1E-8
Expected scale	about 100 m

Integrand program

```
01 LBL "CVSAT"
02 MVAR "T"
03 MVAR "V"
04 RCL "V"
05 END
```

Program-entry keys

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

Line 01: [PGM.FCN] {LBL} {CVSAT} [ENTER]

Line 02: [PGM.FCN] [▲] {MVAR} {T} [ENTER]

Line 03: [PGM.FCN] [▲] {MVAR} {V} [ENTER]

Line 04: [RCL] {V} [ENTER]

The new space already ends in .END.; finish: [PRGM]

Run $\int f(x)$

[f(x)] {CVSAT} → [1] [2] {V} → {T} → [0] {LLIM} → [8] {ULIM} → [1] [E] [+/-] [8] {ACC} → {f}

Read X for the integral. Press [X<>Y] to inspect Y, then [X<>Y] again to restore X.

Register / check	Value
X — integral	96 m

Register / check	Value
Y — returned convergence estimate	approximately 0 m
Independent reference	96 m; $ \Delta \approx 0$ m

VERIFICATION 96 m; $|\Delta| \approx 0$ m. The modeled Y value is an algorithmic estimate from the current Free42 3.3.10 loop, not a proof that the model or data are correct.

KEY IDEA The unit product (m/s)·s predicts metres before calculation.

13.2 Distance under linearly increasing velocity

Start at 3 m/s and accelerate at 2 m/s² for 5 s.

$$s = \int_0^5 (V_0 + AT) \, dT$$

Variable	Meaning	Unit
T	time	s
V0	initial velocity	m/s
A	acceleration	m/s ²

Item	Value
Integration variable	T
LLIM	0
ULIM	5
ACC	1E−8
Expected scale	tens of metres

Integrand program

```

01 LBL "VLIN"
02 MVAR "T"
03 MVAR "V0"
04 MVAR "A"
05 RCL "T"
06 RCL× "A"
07 RCL+ "V0"
08 END

```

Program-entry keys

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)
 Line 01: [PGM.FCN] {LBL} {VLIN} [ENTER]
 Line 02: [PGM.FCN] [▲] {MVAR} {T} [ENTER]

Line 03: [PGM.FCN] [▲] {MVAR} {V0} [ENTER]

Line 04: [PGM.FCN] [▲] {MVAR} {A} [ENTER]

Line 05: [RCL] {T} [ENTER]

Line 06: [RCL] [×] {A} [ENTER]

Line 07: [RCL] [+] {V0} [ENTER]

The new space already ends in .END.; finish: [PRGM]

Run $\int f(x)$

[$\int f(x)$] {VLIN} → [3] {V0} → [2] {A} → {T} → [0] {LLIM} → [5] {ULIM} → [1] [E] [+/-] [8] {ACC} → { $\int$ }

Read X for the integral. Press [X<>Y] to inspect Y, then [X<>Y] again to restore X.

Register / check	Value
X — integral	40 m
Y — returned convergence estimate	approximately 0 m
Independent reference	40 m; $ \Delta \approx 0$ m

VERIFICATION 40 m; $|\Delta| \approx 0$ m. The modeled Y value is an algorithmic estimate from the current Free42 3.3.10 loop, not a proof that the model or data are correct.

KEY IDEA Parameters remain fixed while T is varied by the integrator.

13.3 Work by a constant force

A 50 N force acts through 3 m.

$$W = \int_0^3 F \, dx$$

Variable	Meaning	Unit
X	displacement	m
F	force	N

Item	Value
Integration variable	X
LLIM	0
ULIM	3
ACC	1E-8
Expected scale	150 J

Integrand program

```

01 LBL "CFORC"
02 MVAR "X"
03 MVAR "F"
04 RCL "F"
05 END

```

Program-entry keys

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

Line 01: [PGM.FCN] {LBL} {CFORC} [ENTER]

Line 02: [PGM.FCN] [▲] {MVAR} {X} [ENTER]

Line 03: [PGM.FCN] [▲] {MVAR} {F} [ENTER]

Line 04: [RCL] {F} [ENTER]

The new space already ends in .END.; finish: [PRGM]

Run $\int f(x)$

[$\int$ f(x)] {CFORC} → [5] [0] {F} → {X} → [0] {LLIM} → [3] {ULIM} → [1] [E] [+/-] [8] {ACC} → { $\int$ }

Read X for the integral. Press [X<>Y] to inspect Y, then [X<>Y] again to restore X.

Register / check	Value
X — integral	150 J
Y — returned convergence estimate	approximately 0 J
Independent reference	150 J; $ \Delta \approx 0$ J

VERIFICATION 150 J; $|\Delta| \approx 0$ J. The modeled Y value is an algorithmic estimate from the current Free42 3.3.10 loop, not a proof that the model or data are correct.

KEY IDEA N·m = J is a fast units check.

13.4 Work by a linearly varying force

Use $F(x)=20+4x$ N over $0 \leq x \leq 3$ m.

$$W = \int_0^3 (F_0 + KX) dX$$

Variable	Meaning	Unit
X	displacement	m
F0	force at x=0	N
K	force gradient	N/m

Item	Value
Integration variable	X

Item	Value
LLIM	0
ULIM	3
ACC	1E-8
Expected scale	between 60 and 96 J

Integrand program

```

01 LBL "VFORC"
02 MVAR "X"
03 MVAR "F0"
04 MVAR "K"
05 RCL "X"
06 RCL× "K"
07 RCL+ "F0"
08 END

```

Program-entry keys

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

Line 01: [PGM.FCN] {LBL} {VFORC} [ENTER]

Line 02: [PGM.FCN] [▲] {MVAR} {X} [ENTER]

Line 03: [PGM.FCN] [▲] {MVAR} {F0} [ENTER]

Line 04: [PGM.FCN] [▲] {MVAR} {K} [ENTER]

Line 05: [RCL] {X} [ENTER]

Line 06: [RCL] [×] {K} [ENTER]

Line 07: [RCL] [+] {F0} [ENTER]

The new space already ends in .END.; finish: [PRGM]

Run $\int f(x)$

[$\int f(x)$] {VFORC} → [2] [0] {F0} → [4] {K} → {X} → [0] {LLIM} → [3] {ULIM} → [1] [E] [+/-] [8] {ACC} → [$\int$]

Read X for the integral. Press [X<>Y] to inspect Y, then [X<>Y] again to restore X.

Register / check	Value
X — integral	78 J
Y — returned convergence estimate	approximately 0 J
Independent reference	78 J; $ \Delta \approx 0$ J

VERIFICATION 78 J; $|\Delta| \approx 0$ J. The modeled Y value is an algorithmic estimate from the current Free42 3.3.10 loop, not a proof that the model or data are correct.

KEY IDEA Bound the answer between minimum force $\times$ distance and maximum force $\times$ distance.

13.5 Charge from constant current

A current of 0.2 A flows for 30 s.

$$Q = \int_0^{30} I \, dt$$

Variable	Meaning	Unit
T	time	s
I	current	A

Item	Value
Integration variable	T
LLIM	0
ULIM	30
ACC	1E-8
Expected scale	6 C

Integrand program

```

01 LBL "CURNT"
02 MVAR "T"
03 MVAR "I"
04 RCL "I"
05 END

```

Program-entry keys

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

Line 01: [PGM.FCN] {LBL} {CURNT} [ENTER]

Line 02: [PGM.FCN] [▲] {MVAR} {T} [ENTER]

Line 03: [PGM.FCN] [▲] {MVAR} {I} [ENTER]

Line 04: [RCL] {I} [ENTER]

The new space already ends in .END.; finish: [PRGM]

Run $\int f(x)$

[$\int f(x)$] {CURNT} → [0] [.] [2] {I} → {T} → [0] {LLIM} → [3] [0] {ULIM} → [1] [E] [+/-] [8] {ACC} → {f}

Read X for the integral. Press [X<>Y] to inspect Y, then [X<>Y] again to restore X.

Register / check	Value
X — integral	6 C
Y — returned convergence estimate	approximately 0 C
Independent reference	6 C; $ \Delta \approx 0$ C

VERIFICATION 6 C; $|\Delta| \approx 0$ C. The modeled Y value is an algorithmic estimate from the current Free42 3.3.10 loop, not a proof that the model or data are correct.

KEY IDEA $A \cdot s = C$.

13.6 Energy from time-varying power

Power rises from 100 W at 20 W/s for 10 s.

$$E = \int_0^{10} (P_0 + KT) \, dT$$

Variable	Meaning	Unit
T	time	s
P0	initial power	W
K	power slope	W/s

Item	Value
Integration variable	T
LLIM	0
ULIM	10
ACC	1E-8
Expected scale	about 2 kJ

Integrand program

```

01 LBL "POWER"
02 MVAR "T"
03 MVAR "P0"
04 MVAR "K"
05 RCL "T"
06 RCL× "K"
07 RCL+ "P0"
08 END

```

Program-entry keys

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

Line 01: [PGM.FCN] {LBL} {POWER} [ENTER]

Line 02: [PGM.FCN] [▲] {MVAR} {T} [ENTER]

Line 03: [PGM.FCN] [▲] {MVAR} {P0} [ENTER]

Line 04: [PGM.FCN] [▲] {MVAR} {K} [ENTER]

Line 05: [RCL] {T} [ENTER]

Line 06: [RCL] [×] {K} [ENTER]

Line 07: [RCL] [+] {P0} [ENTER]

The new space already ends in .END.; finish: [PRGM]

Run $\int f(x)$

[$\int f(x)$] {POWER} → [1] [0] [0] {P0} → [2] [0] {K} → {T} → [0] {LLIM} → [1] [0] {ULIM} → [1] [E] [+/-] [8] {ACC} → { $\int$ }

Read X for the integral. Press [X<>Y] to inspect Y, then [X<>Y] again to restore X.

Register / check	Value
X — integral	2000 J
Y — returned convergence estimate	approximately 0 J
Independent reference	2000 J; $ \Delta \approx 1.59\text{e-}58$ J

VERIFICATION 2000 J; $|\Delta| \approx 1.59\text{e-}58$ J. The modeled Y value is an algorithmic estimate from the current Free42 3.3.10 loop, not a proof that the model or data are correct.

KEY IDEA A quick trapezoid-area check agrees because the power is linear.

13.7 Probability from a uniform density

X is uniform on [2,6]. Find $P(3 < X < 5)$.

$$P = \int_3^5 1/(6-2) \, dx$$

Variable	Meaning	Unit
X	random value	1
D	constant density	1/unit

Item	Value
Integration variable	X
LLIM	3
ULIM	5
ACC	$1\text{E-}8$
Expected scale	0.5

Integrand program

```
01 LBL "UNIF"
02 MVAR "X"
03 MVAR "D"
04 RCL "D"
05 END
```

Program-entry keys

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

Line 01: [PGM.FCN] {LBL} {UNIF} [ENTER]
 Line 02: [PGM.FCN] [▲] {MVAR} {X} [ENTER]
 Line 03: [PGM.FCN] [▲] {MVAR} {D} [ENTER]
 Line 04: [RCL] {D} [ENTER]

The new space already ends in .END.; finish: [PRGM]

Run $\int f(x)$

[$\int f(x)$] {UNIF} → [0] [.] [2] [5] {D} → {X} → [3] {LLIM} → [5] {ULIM} → [1] [E] [+/-] [8]
 {ACC} → { $\int$ }

Read X for the integral. Press [X<>Y] to inspect Y, then [X<>Y] again to restore X.

Register / check	Value
X — integral	0.5
Y — returned convergence estimate	approximately 0
Independent reference	0.5; $ \Delta \approx 0$

VERIFICATION 0.5; $|\Delta| \approx 0$. The modeled Y value is an algorithmic estimate from the current Free42 3.3.10 loop, not a proof that the model or data are correct.

KEY IDEA A valid density integrates to 1 over its full support.

13.8 Continuous income without discounting

A constant flow of 5000 currency units per year lasts 3 years.

$$V = \int_0^3 C \, dt$$

Variable	Meaning	Unit
T	time	year
C	income rate	currency/year

Item	Value
Integration variable	T
LLIM	0
ULIM	3
ACC	1E-8
Expected scale	15000

Integrand program

01 LBL "INCOM"

```

02 MVAR "T"
03 MVAR "C"
04 RCL "C"
05 END

```

Program-entry keys

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

Line 01: [PGM.FCN] {LBL} {INCOM} [ENTER]

Line 02: [PGM.FCN] [▲] {MVAR} {T} [ENTER]

Line 03: [PGM.FCN] [▲] {MVAR} {C} [ENTER]

Line 04: [RCL] {C} [ENTER]

The new space already ends in .END.; finish: [PRGM]

Run $\int f(x)$

[$\int f(x)$] {INCOM} → [5] [0] [0] [0] {C} → {T} → [0] {LLIM} → [3] {ULIM} → [1] [E] [+/-] [8] {ACC} → { $\int$ }

Read X for the integral. Press [X<>Y] to inspect Y, then [X<>Y] again to restore X.

Register / check	Value
X — integral	15000 currency
Y — returned convergence estimate	approximately 0 currency
Independent reference	15000 currency; $ \Delta \approx 0$ currency

VERIFICATION 15000 currency; $|\Delta| \approx 0$ currency. The modeled Y value is an algorithmic estimate from the current Free42 3.3.10 loop, not a proof that the model or data are correct.

KEY IDEA A cash-flow rate is the integrand; currency itself is the accumulated result.

PART VI

PARAMETERS AND REUSABLE INTEGRANDS

Separate the variable being swept from the constants that define the model.

14 MVAR turns a formula into a reusable numerical model

In SPRNG, X is selected as the integration variable and K is stored as a constant parameter. The limits—not another MVAR—supply the compression distance. Change K and ULIM, then press $\int$ again. The program itself does not change.

14.1 Reusable spring work

For $k=250$ N/m, compress a spring from 0 to 0.08 m.

$$W = \int_0^D KX \, dX$$

Variable	Meaning	Unit
X	displacement	m
K	spring constant	N/m

Item	Value
Integration variable	X
LLIM	0
ULIM	0.08
ACC	1E-8
Expected scale	less than 1 J

Integrand program

```

01 LBL "SPRNG"
02 MVAR "X"
03 MVAR "K"
04 RCL "X"
05 RCL× "K"
06 END

```

Program-entry keys

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

Line 01: [PGM.FCN] {LBL} {SPRNG} [ENTER]

Line 02: [PGM.FCN] [▲] {MVAR} {X} [ENTER]

Line 03: [PGM.FCN] [▲] {MVAR} {K} [ENTER]

Line 04: [RCL] {X} [ENTER]

Line 05: [RCL] [×] {K} [ENTER]

The new space already ends in .END.; finish: [PRGM]

Run $\int f(x)$

[$\int f(x)$] {SPRNG} → [2] [5] [0] {K} → {X} → [0] {LLIM} → [0] [.] [0] [8] {ULIM} → [1] [E]
 [+/-] [8] {ACC} → { $\int$ }

Read X for the integral. Press [X<>Y] to inspect Y, then [X<>Y] again to restore X.

Register / check	Value
X — integral	0.8 J
Y — returned convergence estimate	approximately 0 J
Independent reference	0.8 J; $ \Delta \approx 0$ J

VERIFICATION 0.8 J; $|\Delta| \approx 0$ J. The modeled Y value is an algorithmic estimate from the current Free42 3.3.10 loop, not a proof that the model or data are correct.

KEY IDEA Change K or ULIM without rewriting the integrand.

K (N/m)	d (m)	Expected $W = Kd^2/2$ (J)	What changes on DM42
100	0.05	0.125	store K; set ULIM
250	0.08	0.800	store K; set ULIM
500	0.10	2.500	store K; set ULIM
250	0.16	3.200	only ULIM changes; doubling d quadruples W

EXPERT TIP Use descriptive parameters but keep names short. MVAR declarations should remain together after the global label so the integration application's variable menu is predictable.

14.2 Reuse patterns across disciplines

Model	Integration variable	Parameters that stay fixed	Typical reuse
$F_0 e^{(-t/\tau)}$	T	F0, TAU	different impulse duration or decay time
$C_0 e^{(-kt)}$	T	C0, K	different AUC horizon or rate
$C(1+gt)e^{(-rt)}$	T	C, G, R	cash-flow scenarios
normal density	Z	MU, SIG if generalized	probability intervals

PART VII

INTERMEDIATE EXAMPLES

Use integration where it starts to earn its keep across science, engineering and decisions.

15 A balanced working library

Each program below is still shown with complete entry guidance. The analytical checks are retained where possible; the Gaussian, Arrhenius and probability examples begin the transition to integrals whose elementary antiderivatives are inconvenient or absent.

15.1 Impulse from a decaying force

A force $F(t)=100e^{(-t/2)}$ N acts for 6 s.

$$J = \int_0^6 F_0 e^{(-T/\tau)} dT$$

Variable	Meaning	Unit
T	time	s
F0	initial force	N
TAU	decay time	s

Item	Value
Integration variable	T
LLIM	0
ULIM	6
ACC	1E-9
Expected scale	slightly below 200 N·s

Integrand program

```

01 LBL "IMPUL"
02 MVAR "T"
03 MVAR "F0"
04 MVAR "TAU"
05 RCL "T"
06 +/-
07 RCL÷ "TAU"
08 E↑X
09 RCL× "F0"
10 END

```

Program-entry keys

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

Line 01: [PGM.FCN] {LBL} {IMPUL} [ENTER]

Line 02: [PGM.FCN] [▲] {MVAR} {T} [ENTER]

Line 03: [PGM.FCN] [▲] {MVAR} {F0} [ENTER]

Line 04: [PGM.FCN] [▲] {MVAR} {TAU} [ENTER]

Line 05: [RCL] {T} [ENTER]

Line 06: [+/-]

Line 07: [RCL] [÷] (TAU) [ENTER]

Line 08: [E+X]

Line 09: [RCL] [×] (F0) [ENTER]

The new space already ends in .END.; finish: [PRGM]

Run ∫f(x)

[∫f(x)] {IMPUL} → [1] [0] [0] {F0} → [2] {TAU} → {T} → [0] {LLIM} → [6] {ULIM} → [1] [E]
[+/-] [9] {ACC} → {f}

Read X for the integral. Press [X<>Y] to inspect Y, then [X<>Y] again to restore X.

Register / check	Value
X — integral	190.042586326 N·s
Y — returned convergence estimate	approximately 1.44e-08 N·s
Independent reference	190.042586326 N·s; Δ ≈ 4.09e-12 N·s

VERIFICATION 190.042586326 N·s; |Δ| ≈ 4.09e-12 N·s. The modeled Y value is an algorithmic estimate from the current Free42 3.3.10 loop, not a proof that the model or data are correct.

KEY IDEA A decaying tail can be bounded by the omitted exponential remainder.

15.2 Distance from nonlinear velocity

A model $v(t)=4+0.8t^2$ m/s is used for 5 s.

$$s = \int_0^5 (V_0 + AT^2) dT$$

Variable	Meaning	Unit
T	time	s
V0	base velocity	m/s
A	quadratic coefficient	m/s ³

Item	Value
Integration variable	T
LLIM	0
ULIM	5
ACC	1E-9
Expected scale	about 50 m

Integrand program

```

01 LBL "DIST"
02 MVAR "T"
03 MVAR "V0"
04 MVAR "A"
05 RCL "T"
06 X↑2
07 RCL× "A"
08 RCL+ "V0"
09 END

```

Program-entry keys

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

Line 01: [PGM.FCN] {LBL} {DIST} [ENTER]

Line 02: [PGM.FCN] [▲] {MVAR} {T} [ENTER]

Line 03: [PGM.FCN] [▲] {MVAR} {V0} [ENTER]

Line 04: [PGM.FCN] [▲] {MVAR} {A} [ENTER]

Line 05: [RCL] {T} [ENTER]

Line 06: [X↑2]

Line 07: [RCL] [×] {A} [ENTER]

Line 08: [RCL] [+] {V0} [ENTER]

The new space already ends in .END.; finish: [PRGM]

Run $\int f(x)$

[f(x)] {DIST} → [4] {V0} → [0] [.] [8] {A} → {T} → [0] {LLIM} → [5] {ULIM} → [1] [E]
 [+/-] [9] {ACC} → {f}

Read X for the integral. Press [X↔Y] to inspect Y, then [X↔Y] again to restore X.

Register / check	Value
X — integral	53.333333333 m
Y — returned convergence estimate	approximately 0 m
Independent reference	53.333333333 m; $ \Delta \approx 0$ m

VERIFICATION 53.333333333 m; $|\Delta| \approx 0$ m. The modeled Y value is an algorithmic estimate from the current Free42 3.3.10 loop, not a proof that the model or data are correct.

KEY IDEA The units of A must be m/s^3 so that AT^2 is a velocity.

15.3 Energy in a resistor during a transient

Current is $i(t) = 2e^{(-t/0.5)}$ A through $10\ \Omega$ for 2 s.

$$E = \int_0^2 I_0^2 e^{(-2T/\tau)} R \, dT$$

Variable	Meaning	Unit
T	time	s
I_0	initial current	A

Variable	Meaning	Unit
TAU	time constant	s
R	resistance	Ω

Item	Value
Integration variable	T
LLIM	0
ULIM	2
ACC	1E-9
Expected scale	about 10 J

Integrand program

```

01 LBL "RENGY"
02 MVAR "T"
03 MVAR "I0"
04 MVAR "TAU"
05 MVAR "R"
06 RCL "T"
07 2
08 ×
09 +/-
10 RCL÷ "TAU"
11 E+X
12 RCL "I0"
13 X+2
14 ×
15 RCL× "R"
16 END

```

Program-entry keys

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

```

Line 01: [PGM.FCN] {LBL} {RENGY} [ENTER]
Line 02: [PGM.FCN] [▲] {MVAR} {T} [ENTER]
Line 03: [PGM.FCN] [▲] {MVAR} {I0} [ENTER]
Line 04: [PGM.FCN] [▲] {MVAR} {TAU} [ENTER]
Line 05: [PGM.FCN] [▲] {MVAR} {R} [ENTER]
Line 06: [RCL] {T} [ENTER]
Line 07: [2]
Line 08: [×]
Line 09: [+/-]
Line 10: [RCL] [÷] {TAU} [ENTER]
Line 11: [E+X]
Line 12: [RCL] {I0} [ENTER]
Line 13: [X+2]
Line 14: [×]
Line 15: [RCL] [×] {R} [ENTER]

```

The new space already ends in .END.; finish: [PRGM]

Run $\int f(x)$

[$\int f(x)$] {RENGY} → [2] {I0} → [0] [.] [5] {TAU} → [1] [0] {R} → {T} → [0] {LLIM} → [2] {ULIM} → [1] [E] [+/-] [9] {ACC} → {f}

Read X for the integral. Press [X<>Y] to inspect Y, then [X<>Y] again to restore X.

Register / check	Value
X — integral	9.99664537372 J
Y — returned convergence estimate	approximately 1.26e-11 J
Independent reference	9.99664537372 J; $ \Delta \approx 1.89\text{e-}14$ J

VERIFICATION 9.99664537372 J; $|\Delta| \approx 1.89\text{e-}14$ J. The modeled Y value is an algorithmic estimate from the current Free42 3.3.10 loop, not a proof that the model or data are correct.

KEY IDEA Square the current before multiplying by resistance.

15.4 Hydrostatic force on a vertical rectangular gate

A 2 m wide gate extends from the water surface to 3 m depth. Use $\rho=998 \text{ kg/m}^3$.

$$F = \int_0^3 \rho g B Y \, dY$$

Variable	Meaning	Unit
Y	depth	m
RHO	water density	kg/m ³
G	gravity	m/s ²
B	gate width	m

Item	Value
Integration variable	Y
LLIM	0
ULIM	3
ACC	1E-8
Expected scale	about 90 kN

Integrand program

01 LBL "HYDRO"

```

02 MVAR "Y"
03 MVAR "RHO"
04 MVAR "G"
05 MVAR "B"
06 RCL "Y"
07 RCL× "RHO"
08 RCL× "G"
09 RCL× "B"
10 END

```

Program-entry keys

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

Line 01: [PGM.FCN] {LBL} {HYDRO} [ENTER]

Line 02: [PGM.FCN] [▲] {MVAR} {Y} [ENTER]

Line 03: [PGM.FCN] [▲] {MVAR} {RHO} [ENTER]

Line 04: [PGM.FCN] [▲] {MVAR} {G} [ENTER]

Line 05: [PGM.FCN] [▲] {MVAR} {B} [ENTER]

Line 06: [RCL] {Y} [ENTER]

Line 07: [RCL] [×] {RHO} [ENTER]

Line 08: [RCL] [×] {G} [ENTER]

Line 09: [RCL] [×] {B} [ENTER]

The new space already ends in .END.; finish: [PRGM]

Run $f(x)$

[f(x)] {HYDRO} → [9] [9] [8] {RHO} → [9] [.] [8] [0] [6] [6] [5] {G} → [2] {B} → {Y} → [0]
 {LLIM} → [3] {ULIM} → [1] [E] [+/-] [8] {ACC} → {f}

Read X for the integral. Press [X<>Y] to inspect Y, then [X<>Y] again to restore X.

Register / check	Value
X — integral	88083.3303 N
Y — returned convergence estimate	approximately 0 N
Independent reference	88083.3303 N; $ \Delta \approx 1.02\text{e-}56$ N

VERIFICATION 88083.3303 N; $|\Delta| \approx 1.02\text{e-}56$ N. The modeled Y value is an algorithmic estimate from the current Free42 3.3.10 loop, not a proof that the model or data are correct.

KEY IDEA Pressure is zero at the surface and increases linearly with depth.

15.5 Analytically checkable concentration AUC

An educational one-compartment model uses $C(t)=20e^{(-0.25t)}$ mg/L over 24 h.

$$\text{AUC} = \int_0^{24} C_0 e^{(-KT)} dT$$

Variable	Meaning	Unit
T	time	h
C0	initial concentration	mg/L

Variable	Meaning	Unit
K	elimination rate	1/h

Item	Value
Integration variable	T
LLIM	0
ULIM	24
ACC	1E-9
Expected scale	near 80 mg·h/L

Integrand program

```

01 LBL "AUC1"
02 MVAR "T"
03 MVAR "C0"
04 MVAR "K"
05 RCL "T"
06 RCL× "K"
07 +/-
08 E+X
09 RCL× "C0"
10 END

```

Program-entry keys

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

Line 01: [PGM.FCN] {LBL} {AUC1} [ENTER]

Line 02: [PGM.FCN] {▲} {MVAR} {T} [ENTER]

Line 03: [PGM.FCN] {▲} {MVAR} {C0} [ENTER]

Line 04: [PGM.FCN] {▲} {MVAR} {K} [ENTER]

Line 05: [RCL] {T} [ENTER]

Line 06: [RCL] [×] {K} [ENTER]

Line 07: [+/-]

Line 08: [E+X]

Line 09: [RCL] [×] {C0} [ENTER]

The new space already ends in .END.; finish: [PRGM]

Run $\int f(x)$

[f(x)] {AUC1} → [2] [0] {C0} → [0] [.] [2] [5] {K} → {T} → [0] {LLIM} → [2] [4] {ULIM} → [1] [E] [+/-] [9] {ACC} → {f}

Read X for the integral. Press [X<>Y] to inspect Y, then [X<>Y] again to restore X.

Register / check	Value
X — integral	79.8016998259 mg·h/L
Y — returned convergence estimate	approximately 6.03e-12 mg·h/L

Register / check	Value
Independent reference	79.8016998259 mg·h/L; $ \Delta \approx 1.94\text{e-}14$ mg·h/L

VERIFICATION 79.8016998259 mg·h/L; $|\Delta| \approx 1.94\text{e-}14$ mg·h/L. The modeled Y value is an algorithmic estimate from the current Free42 3.3.10 loop, not a proof that the model or data are correct.

KEY IDEA This is a mathematical illustration, not dosing or medical advice.

15.6 Present value of a growing continuous cash flow

$C(t)=10000(1+0.03t)$ per year is discounted continuously at $r=0.05$ over ten years.

$$PV = \int_0^{10} C(1+GT)e^{-RT} dT$$

Variable	Meaning	Unit
T	time	year
C	base flow	currency/year
G	linear growth	1/year
R	discount rate	1/year

Item	Value
Integration variable	T
LLIM	0
ULIM	10
ACC	1E-8
Expected scale	of order 10^5

Integrand program

```

01 LBL "PVAL"
02 MVAR "T"
03 MVAR "C"
04 MVAR "G"
05 MVAR "R"
06 RCL "T"
07 RCL× "G"
08 1
09 +
10 RCL× "C"
11 RCL "T"
12 RCL× "R"
13 +/-
14 E↑X
15 ×

```

16 END

Program-entry keys**Start:** [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

Line 01: [PGM.FCN] {LBL} {PVAL} [ENTER]
 Line 02: [PGM.FCN] [▲] {MVAR} {T} [ENTER]
 Line 03: [PGM.FCN] [▲] {MVAR} {C} [ENTER]
 Line 04: [PGM.FCN] [▲] {MVAR} {G} [ENTER]
 Line 05: [PGM.FCN] [▲] {MVAR} {R} [ENTER]
 Line 06: [RCL] {T} [ENTER]
 Line 07: [RCL] [×] {G} [ENTER]
 Line 08: [1]
 Line 09: [+]
 Line 10: [RCL] [×] {C} [ENTER]
 Line 11: [RCL] {T} [ENTER]
 Line 12: [RCL] [×] {R} [ENTER]
 Line 13: [+/-]
 Line 14: [E↑X]
 Line 15: [×]

The new space already ends in .END.; finish: [PRGM]**Run $f(x)$**

[f(x)] {PVAL} → [1] [0] [0] [0] [0] {C} → [0] [.] [0] [3] {G} → [0] [.] [0] [5] {R} → {T} →
 [0] {LLIM} → [1] [0] {ULIM} → [1] [E] [+/-] [8] {ACC} → {f}

Read X for the integral. Press [X<>Y] to inspect Y, then [X<>Y] again to restore X.

Register / check	Value
X — integral	89518.3493092 currency
Y — returned convergence estimate	approximately 8.12e-06 currency
Independent reference	89518.3493092 currency; $ \Delta \approx 4.11\text{e-}09$ currency

VERIFICATION 89518.3493092 currency; $|\Delta| \approx 4.11\text{e-}09$ currency. The modeled Y value is an algorithmic estimate from the current Free42 3.3.10 loop, not a proof that the model or data are correct.

KEY IDEA Keep growth assumptions and discount convention explicit.

15.7 Probability under a normal densityFind standard-normal probability between $z=-1.96$ and $z=1.96$.

$$P = \int_{-1.96}^{1.96} e^{\widehat{(-Z^2/2)}} / \sqrt{(2\pi)} dZ$$

Variable	Meaning	Unit
Z	standard score	1

Item	Value
Integration variable	Z
LLIM	−1.96
ULIM	1.96
ACC	1E−9
Expected scale	about 0.95

Integrand program

```

01 LBL "NORM"
02 MVAR "Z"
03 RCL "Z"
04 X↑2
05 2
06 ÷
07 +/-
08 E↑X
09 2
10 PI
11 ×
12 SQRT
13 ÷
14 END

```

Program-entry keys

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

Line 01: [PGM.FCN] {LBL} {NORM} [ENTER]

Line 02: [PGM.FCN] [▲] {MVAR} {Z} [ENTER]

Line 03: [RCL] {Z} [ENTER]

Line 04: [X↑2]

Line 05: [2]

Line 06: [÷]

Line 07: [+/-]

Line 08: [E↑X]

Line 09: [2]

Line 10: [π]

Line 11: [×]

Line 12: [√x]

Line 13: [÷]

The new space already ends in .END.; finish: [PRGM]

Run $\int f(x)$

[f(x)] {NORM} → {Z} → [1] [.] [9] [6] [+/-] {LLIM} → [1] [.] [9] [6] {ULIM} → [1] [E] [+/-]
[9] {ACC} → {f}

Read X for the integral. Press [X<>Y] to inspect Y, then [X<>Y] again to restore X.

Register / check	Value
X — integral	0.950004209704

Register / check	Value
Y — returned convergence estimate	approximately 6.43e-14
Independent reference	0.950004209704; $ \Delta \approx 3.63\text{e-}17$

VERIFICATION 0.950004209704; $|\Delta| \approx 3.63\text{e-}17$. The modeled Y value is an algorithmic estimate from the current Free42 3.3.10 loop, not a proof that the model or data are correct.

KEY IDEA Probability offers a natural 0-to-1 plausibility check.

15.8 Expected value from a density

For Beta(2,3), $f(x)=12x(1-x)^2$. Integrate $x f(x)$ to find $E[X]$.

$$E[X] = \int_0^1 12X^2(1-X)^2 dX$$

Variable	Meaning	Unit
X	random value	1

Item	Value
Integration variable	X
LLIM	0
ULIM	1
ACC	1E-9
Expected scale	between 0 and 1

Integrand program

```

01 LBL "BMEAN"
02 MVAR "X"
03 RCL "X"
04 X↑2
05 1
06 RCL - "X"
07 X↑2
08 ×
09 12
10 ×
11 END

```

Program-entry keys

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

Line 01: [PGM.FCN] {LBL} {BMEAN} [ENTER]

Line 02: [PGM.FCN] [▲] {MVAR} {X} [ENTER]

Line 03: [RCL] {X} [ENTER]

Line 04: [X↑2]
 Line 05: [1]
 Line 06: [RCL] [-] (X) [ENTER]
 Line 07: [X↑2]
 Line 08: [×]
 Line 09: [1] [2]
 Line 10: [×]

The new space already ends in .END.; finish: [PRGM]

Run ∫f(x)

[∫f(x)] {BMEAN} → {X} → [0] {LLIM} → [1] {ULIM} → [1] [E] [+/-] [9] {ACC} → {∫}

Read X for the integral. Press [X<>Y] to inspect Y, then [X<>Y] again to restore X.

Register / check	Value
X — integral	0.4
Y — returned convergence estimate	approximately 1.87e-10
Independent reference	0.4; Δ ≈ 1.19e-13

VERIFICATION 0.4; |Δ| ≈ 1.19e-13. The modeled Y value is an algorithmic estimate from the current Free42 3.3.10 loop, not a proof that the model or data are correct.

KEY IDEA Expectation multiplies the outcome by its density before integration.

15.9 No convenient elementary antiderivative

Compute ∫ from −1 to 1 of e^{−x²}.

$$I = \int_{-1}^1 e^{-X^2} dX$$

Variable	Meaning	Unit
X	integration variable	1

Item	Value
Integration variable	X
LLIM	−1
ULIM	1
ACC	1E−10
Expected scale	about 1.5

Integrand program

```

01 LBL "GAUSS"
02 MVAR "X"
03 RCL "X"
04 X↑2
05 +/-
06 E↑X
07 END

```

Program-entry keys

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

Line 01: [PGM.FCN] {LBL} {GAUSS} [ENTER]

Line 02: [PGM.FCN] {▲} {MVAR} {X} [ENTER]

Line 03: [RCL] {X} [ENTER]

Line 04: [X↑2]

Line 05: [+/-]

Line 06: [E↑X]

The new space already ends in .END.; finish: [PRGM]

Run $\int f(x)$

[$\int f(x)$] {GAUSS} → {X} → [1] [+/-] {LLIM} → [1] {ULIM} → [1] [E] [+/-] [1] [0] {ACC} → { $\int$ }

Read X for the integral. Press [X<>Y] to inspect Y, then [X<>Y] again to restore X.

Register / check	Value
X — integral	1.49364826562
Y — returned convergence estimate	approximately 7.17e-14
Independent reference	1.49364826562; $ \Delta \approx 8.51\text{e-}17$

VERIFICATION 1.49364826562; $|\Delta| \approx 8.51\text{e-}17$. The modeled Y value is an algorithmic estimate from the current Free42 3.3.10 loop, not a proof that the model or data are correct.

KEY IDEA Numerical integration is especially useful when an elementary antiderivative is unavailable.

Stack construction for $e^{(-X^2)}$

After instruction	T	Z	Y	X
RCL X	—	—	—	x
X↑2	—	—	—	x^2
+/-	—	—	—	$-x^2$
E↑X	—	—	—	$e^{(-x^2)}$

15.10 Variable density and centre of mass: mass

A 3 m rod has linear density $\lambda(x)=2+x$ kg/m. First calculate total mass.

$$M = \int_0^3 (2+X) dX$$

Variable	Meaning	Unit
X	position along rod	m

Item	Value
Integration variable	X
LLIM	0
ULIM	3
ACC	1E-9
Expected scale	about 10 kg

Integrand program

```

01 LBL "RODM"
02 MVAR "X"
03 RCL "X"
04 2
05 +
06 END

```

Program-entry keys

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

Line 01: [PGM.FCN] {LBL} {RODM} [ENTER]

Line 02: [PGM.FCN] [▲] {MVAR} {X} [ENTER]

Line 03: [RCL] {X} [ENTER]

Line 04: [2]

Line 05: [+]

The new space already ends in .END.; finish: [PRGM]

Run $\int f(x)$

[$\int f(x)$] {RODM} → {X} → [0] {LLIM} → [3] {ULIM} → [1] [E] [+/-] [9] {ACC} → {f}

Read X for the integral. Press [X<>Y] to inspect Y, then [X<>Y] again to restore X.

Register / check	Value
X — integral	10.5 kg
Y — returned convergence estimate	approximately 0 kg
Independent reference	10.5 kg; $ \Delta \approx 0$ kg

VERIFICATION 10.5 kg; $|\Delta| \approx 0$ kg. The modeled Y value is an algorithmic estimate from the current Free42 3.3.10 loop, not a proof that the model or data are correct.

KEY IDEA Centre of mass requires both mass and first moment; do not integrate a ratio point by point.

15.11 Variable density and centre of mass: first moment

Reuse the same density in $x\lambda(x)$, then divide first moment by mass.

$$N = \int_0^3 X(2+X)dX; \quad \bar{x}=N/M=1.714285... \text{ m}$$

Variable	Meaning	Unit
X	position along rod	m

Item	Value
Integration variable	X
LLIM	0
ULIM	3
ACC	1E-9
Expected scale	about 18 kg·m

Integrand program

```

01 LBL "RODN"
02 MVAR "X"
03 RCL "X"
04 2
05 RCL+ "X"
06 ×
07 END

```

Program-entry keys

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

Line 01: [PGM.FCN] {LBL} {RODN} [ENTER]

Line 02: [PGM.FCN] [▲] {MVAR} {X} [ENTER]

Line 03: [RCL] {X} [ENTER]

Line 04: [2]

Line 05: [RCL] [+] {X} [ENTER]

Line 06: [×]

The new space already ends in .END.; finish: [PRGM]

Run $\int f(x)$

[f(x)] {RODN} → {X} → [0] {LLIM} → [3] {ULIM} → [1] [E] [+/-] [9] {ACC} → {f}

Read X for the integral. Press [X<>Y] to inspect Y, then [X<>Y] again to restore X.

Register / check	Value
X — integral	18 kg·m

Register / check	Value
Y — returned convergence estimate	approximately 0 kg·m
Independent reference	18 kg·m; $ \Delta \approx 0$ kg·m

VERIFICATION 18 kg·m; $|\Delta| \approx 0$ kg·m. The modeled Y value is an algorithmic estimate from the current Free42 3.3.10 loop, not a proof that the model or data are correct.

KEY IDEA The ratio of two verified integrals produces the centroid. Units reduce from kg·m/kg to m.

15.12 Temperature-ramped reaction exposure

For $k(T)=A \exp[-E_a/(RT)]$ and $T(t)=300+5t$ K, integrate k over 20 s using $A=1\text{E}7 \text{ s}^{-1}$ and $E_a=80000$ J/mol.

$$K = \int_0^{20} A \exp\{-E_a/[R(T_0+BT)]\} dT$$

Variable	Meaning	Unit
T	time	s
A	pre-exponential factor	1/s
EA	activation energy	J/mol
R	gas constant	J/(mol·K)
T0	initial temperature	K
B	ramp	K/s

Item	Value
Integration variable	T
LLIM	0
ULIM	20
ACC	1E-8
Expected scale	small but dimensionless

Integrand program

```

01 LBL "ARRH"
02 MVAR "T"
03 MVAR "A"
04 MVAR "EA"
05 MVAR "R"
```

```

06 MVAR "T0"
07 MVAR "B"
08 RCL "T"
09 RCL× "B"
10 RCL+ "T0"
11 RCL× "R"
12 RCL "EA"
13 X<>Y
14 ÷
15 +/-
16 E↑X
17 RCL× "A"
18 END

```

Program-entry keys

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

```

Line 01: [PGM.FCN] {LBL} {ARRH} [ENTER]
Line 02: [PGM.FCN] [▲] {MVAR} {T} [ENTER]
Line 03: [PGM.FCN] [▲] {MVAR} {A} [ENTER]
Line 04: [PGM.FCN] [▲] {MVAR} {EA} [ENTER]
Line 05: [PGM.FCN] [▲] {MVAR} {R} [ENTER]
Line 06: [PGM.FCN] [▲] {MVAR} {T0} [ENTER]
Line 07: [PGM.FCN] [▲] {MVAR} {B} [ENTER]
Line 08: [RCL] {T} [ENTER]
Line 09: [RCL] [×] {B} [ENTER]
Line 10: [RCL] [+] {T0} [ENTER]
Line 11: [RCL] [×] {R} [ENTER]
Line 12: [RCL] {EA} [ENTER]
Line 13: [X<>Y]
Line 14: [÷]
Line 15: [+/-]
Line 16: [E↑X]
Line 17: [RCL] [×] {A} [ENTER]

```

The new space already ends in .END.; finish: [PRGM]

Run $\int f(x)$

```

[∫f(x)] {ARRH} → [1] [E] [7] {A} → [8] [0] [0] [0] [0] {EA} → [8] [.] [3] [1] [4] [4] [6] [2]
[6] [1] [8] {R} → [3] [0] [0] {T0} → [5] {B} → {T} → [0] {LLIM} → [2] [0] {ULIM} → [1] [E]
[+/-] [8] {ACC} → {∫}

```

Read X for the integral. Press [X<>Y] to inspect Y, then [X<>Y] again to restore X.

Register / check	Value
X — integral	0.0011005260494
Y — returned convergence estimate	approximately 1.43e-12
Independent reference	0.0011005260494; $ \Delta \approx 1.61\text{e-}16$

VERIFICATION 0.0011005260494; $|\Delta| \approx 1.61\text{e-}16$. The modeled Y value is an algorithmic estimate from the current Free42 3.3.10 loop, not a proof that the model or data are correct.

What can go wrong

- Using °C directly in an Arrhenius denominator is physically wrong.

KEY IDEA Exponent signs and kelvin units dominate this calculation.

15.13 Expected shortage under uniform demand

Demand is uniform from 80 to 120 units. For stock $Q=100$, integrate the shortage only where $D>Q$.

$$E[(D-Q)^+] = \int_Q^{120} (D-Q)/40 \, dD$$

Variable	Meaning	Unit
D	demand and integration variable	unit
Q	stock quantity	unit
LO	support lower bound	unit
HI	support upper bound	unit

Item	Value
Integration variable	D
LLIM	100
ULIM	120
ACC	1E-9
Expected scale	a few units

Integrand program

```

01 LBL "SHORT"
02 MVAR "D"
03 MVAR "Q"
04 MVAR "LO"
05 MVAR "HI"
06 RCL "D"
07 RCL - "Q"
08 RCL "HI"
09 RCL - "LO"
10 ÷
11 END

```

Program-entry keys

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

```

Line 01: [PGM.FCN] {LBL} {SHORT} [ENTER]
Line 02: [PGM.FCN] [▲] {MVAR} {D} [ENTER]
Line 03: [PGM.FCN] [▲] {MVAR} {Q} [ENTER]
Line 04: [PGM.FCN] [▲] {MVAR} {LO} [ENTER]
Line 05: [PGM.FCN] [▲] {MVAR} {HI} [ENTER]
Line 06: [RCL] {D} [ENTER]
Line 07: [RCL] [-] {Q} [ENTER]
Line 08: [RCL] {HI} [ENTER]
Line 09: [RCL] [-] {LO} [ENTER]

```

Line 10: [÷]

The new space already ends in .END.; finish: [PRGM]

Run $\int f(x)$

[$\int f(x)$] {SHORT} → [1] [0] [0] {Q} → [8] [0] {L0} → [1] [2] [0] {HI} → {D} → [1] [0] [0] {LLIM} → [1] [2] [0] {ULIM} → [1] [E] [+/-] [9] {ACC} → { $\int$ }

Read X for the integral. Press [X<>Y] to inspect Y, then [X<>Y] again to restore X.

Register / check	Value
X — integral	5 unit
Y — returned convergence estimate	approximately 0 unit
Independent reference	5 unit; $ \Delta \approx 0$ unit

VERIFICATION 5 unit; $|\Delta| \approx 0$ unit. The modeled Y value is an algorithmic estimate from the current Free42 3.3.10 loop, not a proof that the model or data are correct.

KEY IDEA Piecewise expectations are clearer and safer when the overage and shortage intervals are integrated separately.

15.14 Expected payoff under continuous uncertainty

An uncertain state X is uniform on [0,1]. Payoff is $U(X)=50+20X-30X^2$.

$$E[U] = \int_0^1 (50+20X-30X^2) dX$$

Variable	Meaning	Unit
X	uncertain state	1

Item	Value
Integration variable	X
LLIM	0
ULIM	1
ACC	1E-9
Expected scale	around 50

Integrand program

```
01 LBL "PAYOF"
02 MVAR "X"
03 RCL "X"
04 20
05 ×
06 50
```

```

07  +
08  RCL "X"
09  X+2
10  30
11  ×
12  −
13  END

```

Program-entry keys

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

Line 01: [PGM.FCN] {LBL} {PAYOF} [ENTER]

Line 02: [PGM.FCN] [▲] {MVAR} {X} [ENTER]

Line 03: [RCL] {X} [ENTER]

Line 04: [2] [0]

Line 05: [×]

Line 06: [5] [0]

Line 07: [+]

Line 08: [RCL] {X} [ENTER]

Line 09: [X+2]

Line 10: [3] [0]

Line 11: [×]

Line 12: [−]

The new space already ends in .END.; finish: [PRGM]

Run $\int f(x)$

[$\int f(x)$] {PAYOF} → {X} → [0] {LLIM} → [1] {ULIM} → [1] [E] [+/-] [9] {ACC} → {f}

Read X for the integral. Press [X<>Y] to inspect Y, then [X<>Y] again to restore X.

Register / check	Value
X — integral	50 payoff
Y — returned convergence estimate	approximately 0 payoff
Independent reference	50 payoff; $ \Delta \approx 0$ payoff

VERIFICATION 50 payoff; $|\Delta| \approx 0$ payoff. The modeled Y value is an algorithmic estimate from the current Free42 3.3.10 loop, not a proof that the model or data are correct.

KEY IDEA A continuous mixed strategy or uncertain state needs a density; here the uniform density equals 1.

PART VIII

SIGN, SYMMETRY AND CANCELLATION

A small answer may be the difference between large contributions.

16 Signed integrals

```

01 LBL "ODD"
02 MVAR "X"
03 RCL "X"
04 END

01 LBL "ABSX"
02 MVAR "X"
03 RCL "X"
04 ABS
05 END

```

Integral	X	Y (modeled)	Meaning
$\int_{-1}^1 x \, dx$	0	approximately 0	positive and negative contributions cancel
$\int_{-1}^1 x \, dx$	1	approximately 0	geometric area is retained

$$\text{odd } f: \int_{-a}^a f(x)dx=0 \quad \text{even } f: \int_{-a}^a f(x)dx=2 \int_0^a f(x)dx$$

Symmetry is both mathematics and numerical strategy. If a function is known to be odd over exactly symmetric bounds, the integral is exactly zero without spending calculator time. If it is even, integrating one half and doubling removes redundant work and can avoid cancellation.

For $\int_0^{2\pi} \sin x \, dx$, integrating the positive half-cycle and negative half-cycle separately exposes the two contributions +2 and -2. Reporting only the final zero hides the scale of cancellation.

NUMERICAL TRAP When $|I|$ is tiny but $\int|f|$ is large, relative accuracy of I can be poor even if both lobes are individually accurate. Track absolute error and calculate lobes separately.

PART IX

INTERVAL SPLITTING

Split where the mathematics changes, where the graph bends, or where verification needs local evidence.

17 Why splitting is principled

$$\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$$

Clue	Useful split
piecewise definition	every breakpoint
jump discontinuity	at the jump; integrate one-sided pieces
narrow peak	bracket the peak with several peak widths
rapid transition	before, through and after the transition
sign change / cancellation	at zeros or lobe boundaries
long tail	core region plus successive tail bands
integrable endpoint singularity	use a substitution, then split if needed

For a piecewise constant example $f(x)=1$ on $[0,1)$ and 100 on $[1,2]$, the mathematically transparent computation is $\int_0^1 1dx + \int_1^2 100dx = 101$. One whole-interval request asks an adaptive sampler to discover the jump; two requests tell it what you already know.

VERIFICATION Compare whole and split results at the same ACC. Agreement is useful evidence. Disagreement identifies a region that needs more attention; it is not an invitation to average the answers.

18 Split-sum worksheet

Piece	LLIM	ULIM	X result	Y estimate	Sign / note
1					
2					
3					
4					
sum	—	—		combine absolute numerical estimates conservatively	

PART X

DISCONTINUITIES AND SINGULARITIES

A calculator can return a number even when the question is badly posed.

19 Classify before calculating

Feature	Example	Ordinary integral?	DM42 strategy
removable discontinuity	$\sin x/x$ at $x=0$	yes after defining limiting value	rewrite or split; do not force 0/0
jump	step function	yes on finite interval	split at jump
endpoint integrable singularity	$1/\sqrt{x}$ on $(0,1]$	yes, value 2	substitute $x=u^2$
endpoint logarithm	$\ln x$ on $(0,1]$	yes, value -1	endpoint avoiding helps; substitution/check still wise
internal pole	$1/(x-c)$	generally no ordinary integral across c	do not integrate through it
nonintegrable endpoint	$1/x$ on $(0,1]$	diverges	stop; change the question

The current integration loop deliberately avoids the requested endpoints. This can make some integrable endpoint singularities numerically approachable, but endpoint avoidance is not a theorem of convergence. An internal singularity can still be sampled directly or distort extrapolation.

$$\int_{-1}^1 1/x \, dx \text{ does not exist as an ordinary improper integral}$$

Symmetric truncation gives a Cauchy principal value of zero, but that is a different definition. The first midpoint sample for symmetric limits is $x=0$, so the current implementation can encounter division by zero immediately. That is a warning from a bad mathematical setup, not a calculator inconvenience to bypass.

DANGER: DISCONTINUITY Never remove or shift a singular sample merely to obtain a display value. Decide whether the intended object is an ordinary integral, a one-sided improper integral, or a principal value, and document that choice.

20 A constructive endpoint substitution

For $\int_0^1 x^{-1/2} dx$, let $x=u^2$. Then $dx=2u \, du$ and $x^{-1/2}=1/u$ for $u>0$, so the transformed integrand is simply 2. The singularity disappears algebraically.

$$\int_0^1 dx/\sqrt{x} = \int_0^1 (1/u)(2u \, du) = \int_0^1 2du = 2$$

```
01 LBL "TWO"
02 MVAR "U"
03 2
04 END
```

WHY THIS WORKS A substitution changes both the function and the differential. Transforming only x and forgetting dx creates the wrong integral.

PART XI

IMPROPER INTEGRALS

Replace infinity with controlled mathematics: truncation, tails, symmetry and tested substitutions.

21 Infinite limits are not an input mode

The DM42 integration menu stores ordinary real LLIM and ULIM values; current documentation does not define an infinity sentinel for these fields. Treat improper integrals by reformulation, not by entering a huge number and hoping it means infinity.

Method 1—finite truncation with a tail bound. For $\int_0^{\infty} e^{-x} dx$, integrate to $B=20$. The omitted tail equals $e^{-20}=2.0611536\text{E}-9$, so the truncated result is 0.999999997938846... and the exact total is 1.

```
01 LBL "DECAY"
02 MVAR "X"
03 RCL "X"
04 +/-
05 E↑X
06 END
```

B	truncated integral $1-e^{-B}$	known tail e^{-B}
5	0.9932620530	6.737947E-3
10	0.9999546001	4.539993E-5
20	0.9999999979	2.061154E-9

Method 2—map infinity to a finite interval. With $x=t/(1-t)$, $dx=dt/(1-t)^2$:

$$\int_0^{\infty} e^{-x} dx = \int_0^1 \exp[-t/(1-t)] / (1-t)^2 dt$$

```
01 LBL "EXPTF"
02 MVAR "T"
03 RCL "T"
04 1
05 RCL - "T"
06 ÷
07 +/-
08 E↑X
09 1
10 RCL - "T"
11 X↑2
12 ÷
13 END
```

The current rule does not evaluate $t=1$ exactly, which is favorable here, but values near 1 combine a rapidly underflowing exponential with a rapidly growing factor. Validate the transformation against the truncation method before reusing it elsewhere.

NUMERICAL TRAP A transformed finite interval can still contain an endpoint singularity, extreme dynamic range or loss of significance. Finite limits alone do not make a problem easy.

PART XII

ADVANCED NUMERICAL BEHAVIOUR

Convergence experiments, scale, peaks, oscillations, round-off and failure diagnosis.

22 The error budget

Error source	What it means	How to probe it
quadrature	finite refinement and extrapolation	tighten ACC; split interval
round-off	finite decimal128 arithmetic	rescale; avoid subtracting near-equals
truncation of an infinite domain	omitted tail	increase cutoff; bound tail
parameter uncertainty	input values are not exact	sensitivity runs
model uncertainty	equations simplify reality	compare models/data
coding error	program differs from intended formula	stack trace; analytic check

23 A convergence experiment

23. Run at a defensible ACC.
24. Record X and Y.
25. Tighten ACC by 100 or 10,000.
26. Record X and Y again.
27. Split at known features and repeat.
28. Compare with an analytic special case or independent computation.
29. Report only the stable, justified digits.

24 Narrow peaks and long flat regions

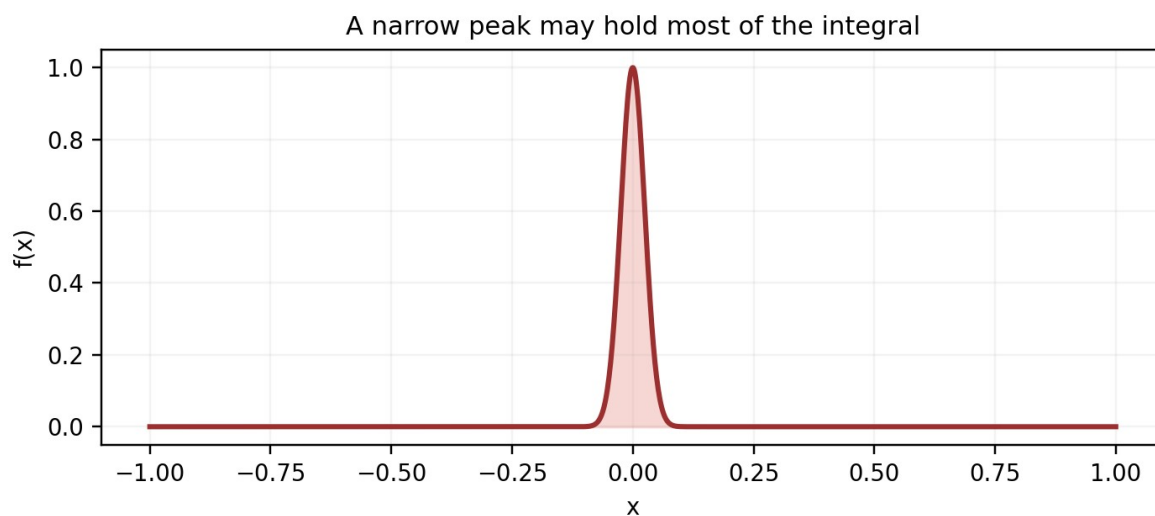


Figure 24.1 A peak much narrower than the full interval can be missed before refinement detects it.

```

01  LBL "NPEAK"
02  MVAR "X"
03  MVAR "C"

```

```

04 MVAR "W"
05 RCL "X"
06 RCL - "C"
07 RCL ÷ "W"
08 X+2
09 +/-
10 E+X
11 END

```

For center C and width W , first integrate a focused region such as $[C-6W, C+6W]$, then the outer pieces if needed. Compare the focused sum with a whole-interval calculation. This does not guarantee success, but it supplies the sampler with scale information you already possess.

25 Oscillation

For $\sin(kx)g(x)$, alternating lobes can cancel and coarse sampling can alias the waveform. Split at zeros or half-periods: $x=n\pi/k$. Sum signed pieces, and separately sum their absolute magnitudes to understand cancellation.

$$\text{half-period width} = \pi/k$$

If $k=50$ on $[0, \pi]$, there are 25 full cycles. A sensible verification uses 50 half-period pieces or groups of known periods rather than one opaque whole-interval request.

26 Deliberate failure cases

Problem	Tempting output	Why it misleads	Correct response
$1/x$ across zero	zero by symmetry	ordinary integral diverges	reject or explicitly define principal value
unresolved narrow peak	almost zero	samples missed dominant region	split around known peak
many oscillations	small number	aliasing/cancellation	split by periods and inspect lobes
e^x on huge interval	overflow	dynamic range exceeds arithmetic	rescale or reformulate
large lobes, tiny difference	many digits in small residual	relative error amplified	integrate/report lobes separately
ACC smaller than data uncertainty	stable long display	model still uncertain	round to defensible significant figures

BEFORE YOU INTEGRATE Plot or sample the integrand mentally or numerically. Locate domain boundaries, sign changes, peaks, discontinuities and natural scales before asking the quadrature algorithm to find them for you.

PART XIII

ADVANCED APPLICATIONS

Derive the quantity, scale it intelligently, run the DM42 and challenge the result.

27 Advanced application map

Field	Integral used	Numerical idea
mechanics / oscillation	finite-amplitude pendulum period	endpoint-near peak as amplitude grows
astrophysics	cosmological distance; distributed mass	dimensionless core + physical scale
electronics	RMS of a harmonic waveform	integrate the square, then take the root
chemistry	Arrhenius exposure under temperature ramp	large exponential sensitivity
biology / medicine	multi-exponential AUC	finite horizon + tail
finance	seasonal growing discounted cash flow	growth, discount and oscillation
operations research	expected excess and shortage	piecewise expectation
decision science	expected payoff under a density	payoff × probability density

The detailed programs are collected below and in Parts XV–XVI. Each follows the same discipline: derive the integrand, list units, predict scale, choose ACC, inspect X/Y, test a split or parameter change, and compare with an independent computation.

28 Finite-amplitude pendulum integrand

Energy conservation gives the exact period in terms of a complete elliptic integral. For amplitude θ_0 , define $k=\sin(\theta_0/2)$. The DM42 integrates only the dimensionless core; the physical scale $4\sqrt{L/g}$ is applied afterward.

$$T = 4 \sqrt{L/g} \int_0^{\pi/2} [1 - k^2 \sin^2 \varphi]^{-1/2} d\varphi$$

```

01 LBL "PEND"
02 MVAR "P"
03 MVAR "K"
04 RCL "P"
05 SIN
06 X↑2
07 RCL "K"
08 X↑2
09 ×
10 1
11 X↔Y
12 −
13 SQRT
14 1/X
15 END

```

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

Line 01: [PGM.FCN] {LBL} {PEND} [ENTER]

Line 02: [PGM.FCN] [▲] {MVAR} {P} [ENTER]

```

Line 03: [PGM.FCN] [▲] {MVAR} {K} [ENTER]
Line 04: [RCL] {P} [ENTER]
Line 05: [SIN]
Line 06: [X↑2]
Line 07: [RCL] {K} [ENTER]
Line 08: [X↑2]
Line 09: [×]
Line 10: [1]
Line 11: [X↔Y]
Line 12: [-]
Line 13: [√x]
Line 14: [1/X]

```

The new space already ends in .END.; finish: [PRGM]

For $\theta_0=120^\circ$, $k=\sin 60^\circ=0.8660254038$. Set RAD, select P, LLIM=0, ULIM= $\pi/2$, ACC= $1\text{E}-10$. The dimensionless X value is then multiplied by $4\sqrt{1/9.80665}$.

Quantity	Value
Integral X	2.1565156475
Y estimate	7.74e-13
Period	2.75456019 s
Small-angle T_0	2.00640929 s

NUMERICAL TRAP As θ_0 approaches 180° , the integrand rises sharply near $\phi=\pi/2$ and the period diverges. Tight ACC cannot rescue an amplitude rounded so close to the limiting case that k is numerically indistinguishable from 1.

29 Nonsinusoidal harmonic waveform RMS

Let $v(t)=2+10\sin(2\pi t)+3\sin(6\pi t)+1.5\cos(10\pi t)$ volts over one second. The waveform is periodic but not a single sine. Integrate v^2 and take the square root of the mean.

$$V^{\text{RMS}} = \sqrt{\int_0^1 v(t)^2 dt / 1 \text{ s}}$$

```

01 LBL "WVRMS"
02 MVAR "T"
03 MVAR "W"
04 RCL "T"
05 RCL× "W"
06 SIN
07 10
08 ×
09 2
10 +
11 RCL "T"
12 RCL× "W"
13 3
14 ×
15 SIN
16 3
17 ×
18 +

```

```

19 RCL "T"
20 RCL× "W"
21 5
22 ×
23 COS
24 1.5
25 ×
26 +
27 X+2
28 END

```

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

```

Line 01: [PGM.FCN] {LBL} {WVRMS} [ENTER]
Line 02: [PGM.FCN] [▲] {MVAR} (T) [ENTER]
Line 03: [PGM.FCN] [▲] {MVAR} (W) [ENTER]
Line 04: [RCL] (T) [ENTER]
Line 05: [RCL] [×] (W) [ENTER]
Line 06: [SIN]
Line 07: [1] [0]
Line 08: [×]
Line 09: [2]
Line 10: [+]
Line 11: [RCL] (T) [ENTER]
Line 12: [RCL] [×] (W) [ENTER]
Line 13: [3]
Line 14: [×]
Line 15: [SIN]
Line 16: [3]
Line 17: [×]
Line 18: [+]
Line 19: [RCL] (T) [ENTER]
Line 20: [RCL] [×] (W) [ENTER]
Line 21: [5]
Line 22: [×]
Line 23: [COS]
Line 24: [1] [.] [5]
Line 25: [×]
Line 26: [+]
Line 27: [X+2]

```

The new space already ends in .END.; finish: [PRGM]

Store $W=2\pi$, select T, integrate 0 to 1 with $ACC=1E-9$. $X=59.625 \text{ V}^2$, then $\sqrt{X}=7.721722\dots \text{ V RMS}$. Y is the estimate for the mean-square integral before the final square root.

Check	Value
Modeled integral X	59.6249999999999999954403 V ²
Modeled Y	4.05216625116e-16 V ²
Independent RMS	7.72172260574 V
Energy in 10 Ω for 1 s	5.9625 J

Orthogonality independently gives mean square $2^2+(10^2+3^2+1.5^2)/2=59.625$. That check is more informative than merely rerunning the same program.

30 Cosmological distance kernels

Assume a spatially flat Λ CDM model with $H_0=67.4 \text{ km}\cdot\text{s}^{-1}\cdot\text{Mpc}^{-1}$, $\Omega_m=0.315$ and $\Omega_\Lambda=0.685$, ignoring radiation for this low-redshift illustration. These are explicit example assumptions, not timeless constants.

$$E(z)=\sqrt{\Omega_m(1+z)^3+\Omega_\Lambda}$$

$$D_c=(c/H_0)\int_0^z dz/E(z); \quad t^L=(1/H_0)\int_0^z dz/[(1+z)E(z)]$$

```

01 LBL "COMOV"
02 MVAR "Z"
03 MVAR "OM"
04 MVAR "OL"
05 RCL "Z"
06 1
07 +
08 3
09 Y↑X
10 RCL× "OM"
11 RCL+ "OL"
12 SQRT
13 1/X
14 END

```

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

```

Line 01: [PGM.FCN] {LBL} {COMOV} [ENTER]
Line 02: [PGM.FCN] [▲] {MVAR} {Z} [ENTER]
Line 03: [PGM.FCN] [▲] {MVAR} {OM} [ENTER]
Line 04: [PGM.FCN] [▲] {MVAR} {OL} [ENTER]
Line 05: [RCL] {Z} [ENTER]
Line 06: [1]
Line 07: [+]
Line 08: [3]
Line 09: [Y↑X]
Line 10: [RCL] [×] {OM} [ENTER]
Line 11: [RCL] [+] {OL} [ENTER]
Line 12: [√x]
Line 13: [1/X]

```

The new space already ends in .END.; finish: [PRGM]

```

01 LBL "LOOKB"
02 MVAR "Z"
03 MVAR "OM"
04 MVAR "OL"
05 RCL "Z"
06 1
07 +
08 3
09 Y↑X
10 RCL× "OM"
11 RCL+ "OL"
12 SQRT
13 1/X
14 RCL "Z"
15 1
16 +

```

```
17 ÷
18 END
```

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

```
Line 01: [PGM.FCN] {LBL} <LOOKB> [ENTER]
Line 02: [PGM.FCN] [▲] {MVAR} <Z> [ENTER]
Line 03: [PGM.FCN] [▲] {MVAR} <OM> [ENTER]
Line 04: [PGM.FCN] [▲] {MVAR} <OL> [ENTER]
Line 05: [RCL] <Z> [ENTER]
Line 06: [1]
Line 07: [+ ]
Line 08: [3]
Line 09: [Y↑X]
Line 10: [RCL] [×] <OM> [ENTER]
Line 11: [RCL] [+ ] <OL> [ENTER]
Line 12: [√x]
Line 13: [1/X]
Line 14: [RCL] <Z> [ENTER]
Line 15: [1]
Line 16: [+ ]
Line 17: [÷ ]
```

The new space already ends in .END.; finish: [PRGM]

For both programs store OM=0.315 and OL=0.685, select Z, and integrate from 0 to 1 with ACC=1E−9. Scale COMOV by c/H₀=4447.9593175 Mpc and LOOKB by 977.79222168/H₀ Gyr.

Output	Dimensionless integral	Scaled result
comoving distance	0.7646794124126391800874369	3401.262917 Mpc
lookback time	0.5480420932950486172175093	7.950613 Gyr

31 Blackbody band kernel

Naive wavelength SI formulas mix powers near 10^{−7} m with exponentials. Substitute $x=hc/(\lambda k^BT)$. The total dimensionless integral is $\pi^4/15$, and a wavelength band maps to reversed x bounds.

$$\text{fraction} = \left[\int_{x_2}^{x_1} x^3 / (e^x - 1) \, dx \right] / (\pi^4/15)$$

```
01 LBL "PLANCK"
02 MVAR "X"
03 RCL "X"
04 3
05 Y↑X
06 RCL "X"
07 E↑X
08 1
09 −
10 ÷
11 END
```

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

```
Line 01: [PGM.FCN] {LBL} <PLANCK> [ENTER]
Line 02: [PGM.FCN] [▲] {MVAR} <X> [ENTER]
```

Line 03: [RCL] (X) [ENTER]
 Line 04: [3]
 Line 05: [Y↑X]
 Line 06: [RCL] (X) [ENTER]
 Line 07: [E↑X]
 Line 08: [1]
 Line 09: [-]
 Line 10: [÷]

The new space already ends in .END.; finish: [PRGM]

At $T=5772$ K and $\lambda=400\text{--}700$ nm, use $LLIM=3.560976333$, $ULIM=6.231708582$, $ACC=1E-9$. The dimensionless band integral is 2.37927011298; divide by $\pi^4/15$.

Quantity	Value
Band fraction	36.638317%
Modeled Y for numerator	6.26258021708e-12
Independent numerator	2.379270112981016640695486

PART XIV

HIGHLY ADVANCED TECHNIQUES

Change variables, normalize scales, decompose intervals and reveal structure before demanding digits.

32 Change of variables as numerical design

Problem	Substitution idea	Benefit	Obligation
huge interval $[a,b]$	$x=a+(b-a)t$	work on $t \in [0,1]$	multiply by $b-a$
endpoint $\sqrt{\cdot}$ singularity	$x=t^2$	removes $1/\sqrt{x}$ behavior	include $dx=2t \, dt$
semi-infinite tail	$x=t/(1-t)$	finite t interval	include $/(1-t)^2$ and inspect $t \rightarrow 1$
physical scale L	$x=L\xi$	dimensionless ξ	factor L outside
spectral Planck law	$x=hc/(\lambda kT)$	moderate dimensionless exponent	reverse bounds and transform $d\lambda$

33 Normalization and dynamic range

Write $I=A\int g(x)dx$ when A contains a large or tiny physical scale and g remains near unity. This separates quadrature conditioning from units. The blackbody and cosmology examples use exactly this architecture.

$$\text{physical result} = \text{physical scale} \times \text{dimensionless integral}$$

34 Decomposition around features

Use narrow subintervals through peaks or transitions and broad subintervals where the function is smooth. For oscillations, align splits with zeros or periods. For piecewise models, split at every branch boundary. For tails, integrate successive bands until their contribution is below the error budget.

EXPERT TIP A good reformulation often improves accuracy more than a smaller ACC. Numerical analysis begins before the $\int$ key is pressed.

35 Parameter sweeps without a programming detour

Keep the program fixed, change one MVAR value, rerun the same limits and record X/Y . Change only one factor at a time unless studying an intentional scenario. Tables in Parts XV–XVI demonstrate amplitude, redshift, order quantity and tail-horizon sweeps.

PART XV

HIGHLY ADVANCED EXAMPLES

*Serious one-dimensional models, explicit assumptions and independently
verified numbers.*

36 Nonlinear pendulum sweep

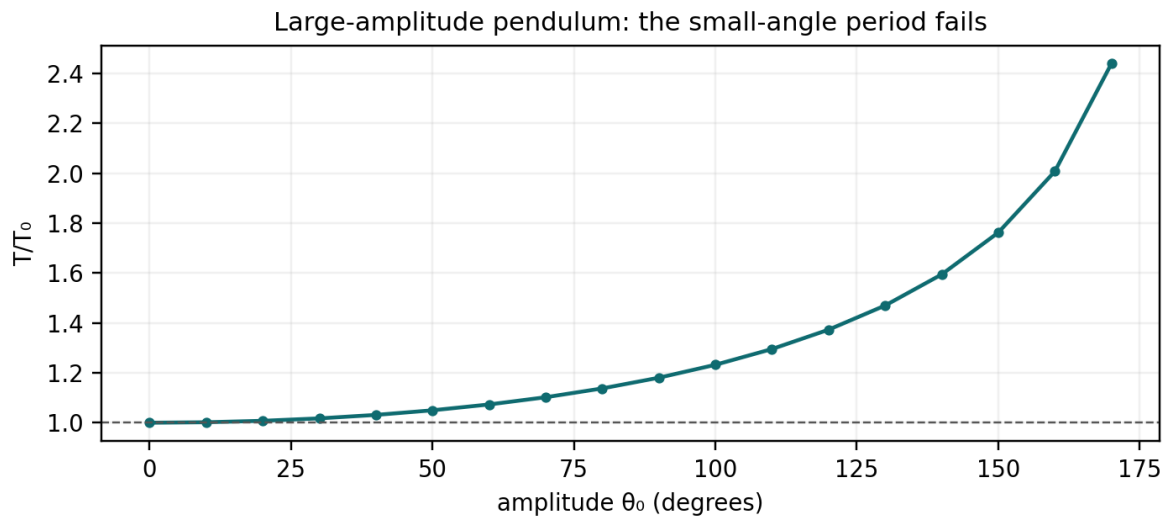


Figure 36.1 Period ratio T/T_0 rises slowly at first, then sharply as θ_0 approaches 180° .

θ_0	k	T/T_0	small-angle error
5°	0.04361938737	1.000476172	0.047617%
30°	0.2588190451	1.017408798	1.7409%
60°	0.5000000000	1.073182007	7.3182%
90°	0.7071067812	1.180340599	18.034%
120°	0.8660254038	1.372880501	37.288%
150°	0.9659258263	1.762203730	76.22%
170°	0.9961946981	2.439362720	143.94%

At 170° the sharp endpoint behavior is visible in the rising ratio. Split the ϕ interval near $\pi/2$ and compare. At exactly 180° , the physical idealization takes infinite time to reach the unstable upright position; there is no finite period to compute.

37 Gaussian probability and tails

For general μ and σ , the normal density is $\exp[-(x-\mu)^2/(2\sigma^2)]/(\sigma\sqrt{2\pi})$. Standardize first when possible: $z=(x-\mu)/\sigma$. The dimensionless standard-normal program in Part VII then needs only transformed limits.

Interval	Probability	Symmetry / tail check
-1 to 1	0.682689492137	central 68.27%

Interval	Probability	Symmetry / tail check
-1.96 to 1.96	0.9500042097035591317268315	central 95.0004%
0 to 3	0.498650101968	half of central $\pm 3\sigma$
3 to 8	0.00134989803163	small tail; scale and absolute error matter

38 Multi-exponential pharmacokinetic exposure

For an educational oral one-compartment model, $C(t) = S[e^{(-k_e t)} - e^{(-k_a t)}]$, where $S = FDk_a/[V(k_a - k_e)]$. Use $F=0.8$, $D=500$ mg, $V=50$ L, $k_a=1.2$ h⁻¹ and $k_e=0.15$ h⁻¹; $S=9.1428571429$ mg/L.

```

01 LBL "AUC2"
02 MVAR "T"
03 MVAR "S"
04 MVAR "KE"
05 MVAR "KA"
06 RCL "T"
07 RCL× "KE"
08 +/-
09 E+X
10 RCL "T"
11 RCL× "KA"
12 +/-
13 E+X
14 -
15 RCL× "S"
16 END

```

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

```

Line 01: [PGM.FCN] {LBL} {AUC2} [ENTER]
Line 02: [PGM.FCN] [▲] {MVAR} {T} [ENTER]
Line 03: [PGM.FCN] [▲] {MVAR} {S} [ENTER]
Line 04: [PGM.FCN] [▲] {MVAR} {KE} [ENTER]
Line 05: [PGM.FCN] [▲] {MVAR} {KA} [ENTER]
Line 06: [RCL] {T} [ENTER]
Line 07: [RCL] [×] {KE} [ENTER]
Line 08: [+/-]
Line 09: [E+X]
Line 10: [RCL] {T} [ENTER]
Line 11: [RCL] [×] {KA} [ENTER]
Line 12: [+/-]
Line 13: [E+X]
Line 14: [-]
Line 15: [RCL] [×] {S} [ENTER]

```

The new space already ends in .END.; finish: [PRGM]

Item	Value
AUC 0-24 h	51.66788739 mg·h/L

Item	Value
Modeled Y	1.19471819558e-8 mg·h/L
AUC 0–∞	53.33333333 mg·h/L
Remaining tail	1.6654459 mg·h/L

CAUTION This is a mathematical integration example, not medical advice. Real pharmacokinetic analysis requires validated models, population variability, assay uncertainty and clinical judgment.

39 Continuous finance with growth and seasonality

Gross flow is $C_0 e^{gt} [1 + a \sin(2\pi t)]$ and present value multiplies by e^{-rt} . Use $C_0 = 100000/\text{year}$, $g = 0.03/\text{year}$, $r = 0.06/\text{year}$, $a = 0.2$ and $T = 5$ years.

```

01 LBL "SEAPV"
02 MVAR "T"
03 MVAR "C"
04 MVAR "G"
05 MVAR "R"
06 MVAR "A"
07 RCL "G"
08 RCL - "R"
09 RCL× "T"
10 E+X
11 RCL× "C"
12 RCL "T"
13 2
14 PI
15 ×
16 ×
17 SIN
18 RCL× "A"
19 1
20 +
21 ×
22 END

```

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

```

Line 01: [PGM.FCN] {LBL} {SEAPV} [ENTER]
Line 02: [PGM.FCN] [▲] {MVAR} {T} [ENTER]
Line 03: [PGM.FCN] [▲] {MVAR} {C} [ENTER]
Line 04: [PGM.FCN] [▲] {MVAR} {G} [ENTER]
Line 05: [PGM.FCN] [▲] {MVAR} {R} [ENTER]
Line 06: [PGM.FCN] [▲] {MVAR} {A} [ENTER]
Line 07: [RCL] {G} [ENTER]
Line 08: [RCL] [-] {R} [ENTER]
Line 09: [RCL] [×] {T} [ENTER]
Line 10: [E+X]
Line 11: [RCL] [×] {C} [ENTER]
Line 12: [RCL] {T} [ENTER]
Line 13: [2]
Line 14: [π]
Line 15: [×]
Line 16: [×]
Line 17: [SIN]

```

Line 18: [RCL] [×] (A) [ENTER]

Line 19: [1]

Line 20: [+]

Line 21: [×]

The new space already ends in .END.; finish: [PRGM]

With T selected, LLIM=0, ULIM=5 and ACC=1E-8, modeled X=464750.1154 currency units and modeled Y≈0.00166. The independent value is 464750.115424.

Gross accumulation is obtained by setting R=0. Comparing gross and discounted totals separates operational growth from time value.

40 Expected inventory cost

Demand D is normal with $\mu=100$ and $\sigma=15$ units. Holding cost is $h=2$ per excess unit and shortage cost $b=8$ per short unit. Integrate expected excess below Q and shortage above Q, truncating at $\mu \pm 6\sigma$ (10 to 190), then form $hE[(Q-D)^+] + bE[(D-Q)^+]$.

Q	E[excess]	E[shortage]	Expected cost
90	2.2667946	12.266795	102.66795
100	5.9841341	5.9841341	59.841341
110	12.266795	2.2667946	42.667946

This is analysis of fixed candidate quantities, not hidden numerical optimization. The three runs show how asymmetric costs can favor carrying more than mean demand.

41 Distributed spherical mass

Let $\rho(r)=5000[1-0.4(r/R)^2]$ kg/m³ with R=1 m. Spherical shells have $dV=4\pi r^2 dr$, so mass is a one-dimensional radial integral.

$$M = \int_0^R 4\pi\rho(r)r^2 dr$$

```

01 LBL "SPMAS"
02 MVAR "R"
03 MVAR "RH0"
04 MVAR "A"
05 RCL "R"
06 X+2
07 RCL "R"
08 X+2
09 RCL× "A"
10 +/-
11 1
12 +
13 ×
14 RCL× "RH0"
15 4
16 ×
17 PI
18 ×
19 END

```

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

```

Line 01: [PGM.FCN] {LBL} (SPMAS) [ENTER]
Line 02: [PGM.FCN] [▲] {MVAR} (R) [ENTER]
Line 03: [PGM.FCN] [▲] {MVAR} (RHO) [ENTER]
Line 04: [PGM.FCN] [▲] {MVAR} (A) [ENTER]
Line 05: [RCL] (R) [ENTER]
Line 06: [X↑2]
Line 07: [RCL] (R) [ENTER]
Line 08: [X↑2]
Line 09: [RCL] [×] (A) [ENTER]
Line 10: [+/-]
Line 11: [1]
Line 12: [+]
Line 13: [×]
Line 14: [RCL] [×] (RHO) [ENTER]
Line 15: [4]
Line 16: [×]
Line 17: [π]
Line 18: [×]

```

The new space already ends in .END.; finish: [PRGM]

Store RHO=5000 and A=0.4; select R; integrate 0 to 1 with ACC=1E-9. $X \approx 15917.40278$ kg, $Y \approx 3.92e-07$ kg. Analytic shell integration independently gives the same result.

PART XVI

EXTREME EXAMPLES

*Six verification dossiers that combine scaling, tails, splitting, sensitivity
and scientific interpretation.*

42 How to read the dossiers

Every dossier states the model, variables, limits, scale, hazards, ACC, program, procedure, X/Y result, independent check and at least one sensitivity test. Constants are examples for numerical study; update them when a real application requires current authoritative values.

43 Extreme 1 — Blackbody band fraction

Problem. What fraction of a 5772 K blackbody's total radiative power lies between 400 and 700 nm? Numerical integration is appropriate because the finite-band Planck kernel has no elementary antiderivative, and dimensionless scaling avoids extreme SI magnitudes.

$$x = hc/(\lambda k^B T); \text{ fraction} = (15/\pi^4) \int_{x_2}^{x_1} x^3/(e^x - 1) dx$$

Item	Definition / value
integration variable	X, dimensionless
limits	3.560976332798569 to 6.231708582397496
parameters	T=5772 K; exact SI h,c,k ^B
expected scale	a substantial fraction, between 0 and 1
hazards	overflow at large x; cancellation in e ^x −1 at tiny x; reversed wavelength bounds
ACC	1E−9, then 1E−11 as a check

```

01 LBL "PLANCK"
02 MVAR "X"
03 RCL "X"
04 3
05 Y↑X
06 RCL "X"
07 E↑X
08 1
09 −
10 ÷
11 END

```

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

Line 01: [PGM.FCN] {LBL} {PLANCK} [ENTER]

Line 02: [PGM.FCN] {▲} {MVAR} {X} [ENTER]

Line 03: [RCL] {X} [ENTER]

Line 04: [3]

Line 05: [Y↑X]

Line 06: [RCL] {X} [ENTER]

Line 07: [E↑X]

Line 08: [1]

Line 09: [−]

Line 10: [÷]

The new space already ends in .END.; finish: [PRGM]

Evidence	Value
X numerator	2.379270112981009600995148
Y estimate	6.26258021708e-12
high-precision fraction	0.366383171384 (36.638317%)
split check	split at midpoint in x; sums agree to reported digits

Sensitivity. Raising temperature moves a fixed wavelength band to different dimensionless limits; the fraction therefore changes even though the universal kernel does not. For very small x , use the calculator's $E \uparrow X - 1$ function if available in the installed core to avoid subtractive loss in $e^x - 1$.

44 Extreme 2 — Cosmological distance and lookback time

Problem. Under the stated flat Λ CDM assumptions, compute comoving distance and lookback time to $z=1$. These are one-dimensional redshift integrals with different kernels, so two programs prevent accidental mixing.

Item	Value
variable	redshift Z , dimensionless
limits	0 to 1
parameters	$H_0=67.4 \text{ km}\cdot\text{s}^{-1}\cdot\text{Mpc}^{-1}$; $\Omega_m=0.315$; $\Omega_\Lambda=0.685$
expected scale	several Gpc and several Gyr
hazards	wrong Hubble-time conversion; omitting $1+z$ in lookback kernel; parameter assumptions
ACC	$1\text{E}-9$

```

01 LBL "COMOV"
02 MVAR "Z"
03 MVAR "OM"
04 MVAR "OL"
05 RCL "Z"
06 1
07 +
08 3
09 Y↑X
10 RCL× "OM"
11 RCL+ "OL"
12 SQRT
13 1/X
14 END

```

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

Line 01: [PGM.FCN] {LBL} {COMOV} [ENTER]

Line 02: [PGM.FCN] [▲] {MVAR} {Z} [ENTER]

Line 03: [PGM.FCN] [▲] {MVAR} {OM} [ENTER]

Line 04: [PGM.FCN] [▲] {MVAR} ⟨OL⟩ [ENTER]
 Line 05: [RCL] ⟨Z⟩ [ENTER]
 Line 06: [1]
 Line 07: [+]
 Line 08: [3]
 Line 09: [Y↑X]
 Line 10: [RCL] [×] ⟨OM⟩ [ENTER]
 Line 11: [RCL] [+] ⟨OL⟩ [ENTER]
 Line 12: [√x]
 Line 13: [1/X]

The new space already ends in .END.; finish: [PRGM]

```
01 LBL "LOOKB"
02 MVAR "Z"
03 MVAR "OM"
04 MVAR "OL"
05 RCL "Z"
06 1
07 +
08 3
09 Y↑X
10 RCL× "OM"
11 RCL+ "OL"
12 SQRT
13 1/X
14 RCL "Z"
15 1
16 +
17 ÷
18 END
```

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

Line 01: [PGM.FCN] {LBL} ⟨LOOKB⟩ [ENTER]
 Line 02: [PGM.FCN] [▲] {MVAR} ⟨Z⟩ [ENTER]
 Line 03: [PGM.FCN] [▲] {MVAR} ⟨OM⟩ [ENTER]
 Line 04: [PGM.FCN] [▲] {MVAR} ⟨OL⟩ [ENTER]
 Line 05: [RCL] ⟨Z⟩ [ENTER]
 Line 06: [1]
 Line 07: [+]
 Line 08: [3]
 Line 09: [Y↑X]
 Line 10: [RCL] [×] ⟨OM⟩ [ENTER]
 Line 11: [RCL] [+] ⟨OL⟩ [ENTER]
 Line 12: [√x]
 Line 13: [1/X]
 Line 14: [RCL] ⟨Z⟩ [ENTER]
 Line 15: [1]
 Line 16: [+]
 Line 17: [÷]

The new space already ends in .END.; finish: [PRGM]

Quantity	X	Y	scaled verification
D _c kernel	0.7646794124126391800874 369	2.03608930545e-14	3401.2629 Mpc
lookback kernel	0.5480420932950486172175 093	9.26121348037e-12	7.9506127 Gyr

Sensitivity. Change Ω_m while preserving flatness ($\Omega_\Lambda = 1 - \Omega_m$) and rerun. The result is model-dependent; the quadrature estimate does not measure cosmological-parameter uncertainty.

45 Extreme 3 — Large-amplitude pendulum

Problem. Compute the period from small amplitude to 170° and quantify failure of the small-angle approximation. The integrand becomes increasingly peaked near $\phi = \pi/2$ as $k \rightarrow 1$.

```

01 LBL "PEND"
02 MVAR "P"
03 MVAR "K"
04 RCL "P"
05 SIN
06 X $\uparrow$ 2
07 RCL "K"
08 X $\uparrow$ 2
09  $\times$ 
10 1
11 X $\leftrightarrow$ Y
12  $-$ 
13 SQRT
14 1/X
15 END

```

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

Line 01: [PGM.FCN] {LBL} (PEND) [ENTER]

Line 02: [PGM.FCN] [$\blacktriangle$] {MVAR} (P) [ENTER]

Line 03: [PGM.FCN] [$\blacktriangle$] {MVAR} (K) [ENTER]

Line 04: [RCL] (P) [ENTER]

Line 05: [SIN]

Line 06: [X $\uparrow$ 2]

Line 07: [RCL] (K) [ENTER]

Line 08: [X $\uparrow$ 2]

Line 09: [$\times$]

Line 10: [1]

Line 11: [X $\leftrightarrow$ Y]

Line 12: [$-$]

Line 13: [$\sqrt{x}$]

Line 14: [1/X]

The new space already ends in .END.; finish: [PRGM]

Required item	Answer
variable / limits	ϕ from 0 to $\pi/2$, RAD mode
parameter	$K = \sin(\theta_0/2)$
physical scale	$4\sqrt{L/g}$; $L = 1$ m, $g = 9.80665$ m/s ²
expected order	near 2 s at small amplitude; larger toward 180°
hazards	endpoint-near peak; K rounding; 180° divergence
ACC strategy	$1\text{E}-8$ then $1\text{E}-10$; split final 10% of ϕ interval

Required item	Answer
X/Y at 120°	2.156515647499643017046303 / 7.73890139071e-13
high-precision T	2.75456019405 s

Interpretation. The familiar $2\pi\sqrt{L/g}$ is a limiting approximation, not the general period. The sensitivity table in §36 is the independent parameter sweep.

46 Extreme 4 — Spectral waveform energy

Problem. For a DC-offset, multi-harmonic waveform, calculate mean square, RMS and energy in 10Ω over one period. Numerical integration remains useful when phase, clipping or additional terms make an orthogonality shortcut inconvenient.

```

01 LBL "WVRMS"
02 MVAR "T"
03 MVAR "W"
04 RCL "T"
05 RCL× "W"
06 SIN
07 10
08 ×
09 2
10 +
11 RCL "T"
12 RCL× "W"
13 3
14 ×
15 SIN
16 3
17 ×
18 +
19 RCL "T"
20 RCL× "W"
21 5
22 ×
23 COS
24 1.5
25 ×
26 +
27 X↑2
28 END

```

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

```

Line 01: [PGM.FCN] {LBL} {WVRMS} [ENTER]
Line 02: [PGM.FCN] [▲] {MVAR} {T} [ENTER]
Line 03: [PGM.FCN] [▲] {MVAR} {W} [ENTER]
Line 04: [RCL] {T} [ENTER]
Line 05: [RCL] [×] {W} [ENTER]
Line 06: [SIN]
Line 07: [1] [0]
Line 08: [×]
Line 09: [2]
Line 10: [+ ]
Line 11: [RCL] {T} [ENTER]
Line 12: [RCL] [×] {W} [ENTER]
Line 13: [3]
Line 14: [×]

```

```
Line 15: [SIN]
Line 16: [3]
Line 17: [×]
Line 18: [+]
```

The new space already ends in .END.; finish: [PRGM]

Item	Value
variable / limits	T, 0 to 1 s
parameter	$W=2\pi$ rad/s
ACC	$1\text{E}-9$
X / Y	59.624999999999999954403 / $4.05216625116\text{e-}16$ V ² ·s
RMS	7.72172260574 V
energy in 10 Ω	5.9625 J
independent check	orthogonal harmonic sum gives 59.625 V ²

Sensitivity and splitting. Split the period into quarters; the signed waveform mean changes by segment, but the nonnegative v^2 pieces must sum to the same energy. If a high-frequency harmonic is added, split finely enough to expose it.

47 Extreme 5 — Pharmacokinetic exposure and remaining tail

Problem. Integrate a two-exponential absorption/elimination concentration model over 24 h, compare with the analytic infinite-horizon AUC, and quantify the remaining tail. Numerical integration is justified for finite windows and extends naturally to more complex models.

```

01  LBL "AUC2"
02  MVAR "T"
03  MVAR "S"
04  MVAR "KE"
05  MVAR "KA"
06  RCL "T"
07  RCL× "KE"
08  +/−
09  E↑X
10  RCL "T"
11  RCL× "KA"
12  +/−

```

```

13  E+X
14  -
15  RCL× "S"
16  END

```

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

```

Line 01: [PGM.FCN] {LBL} {AUC2} [ENTER]
Line 02: [PGM.FCN] {▲} {MVAR} {T} [ENTER]
Line 03: [PGM.FCN] {▲} {MVAR} {S} [ENTER]
Line 04: [PGM.FCN] {▲} {MVAR} {KE} [ENTER]
Line 05: [PGM.FCN] {▲} {MVAR} {KA} [ENTER]
Line 06: [RCL] {T} [ENTER]
Line 07: [RCL] [×] {KE} [ENTER]
Line 08: [+/-]
Line 09: [E+X]
Line 10: [RCL] {T} [ENTER]
Line 11: [RCL] [×] {KA} [ENTER]
Line 12: [+/-]
Line 13: [E+X]
Line 14: [-]
Line 15: [RCL] [×] {S} [ENTER]

```

The new space already ends in .END.; finish: [PRGM]

Item	Value
variable / limits	T, 0 to 24 h
parameters	S=9.1428571429 mg/L; KE=0.15 h ⁻¹ ; KA=1.2 h ⁻¹
expected scale	tens of mg·h/L
hazards	subtracting close exponentials if rates nearly equal; finite tail; model validity
ACC	1E-9
X / Y	51.66788739371220663735113 / 1.19471819558e-8 mg·h/L
independent 0-∞	53.3333333333 mg·h/L

Sensitivity. Repeat with ULIM 12, 24 and 48 h. The increments show whether the chosen observation window captures the slow elimination tail. This remains educational mathematics, not clinical guidance.

48 Extreme 6 — Continuous inventory cost

Problem. For normal demand, calculate excess and shortage expectations separately for fixed Q, combine asymmetric costs, and repeat over candidate quantities. Piecewise decomposition is the numerical idea: each integrand is nonnegative on its own interval.

```

01  LBL "NOVER"
02  MVAR "D"
03  MVAR "Q"
04  MVAR "MU"

```

```

05 MVAR "SIG"
06 RCL "Q"
07 RCL- "D"
08 RCL "D"
09 RCL- "MU"
10 RCL÷ "SIG"
11 X↑2
12 2
13 ÷
14 +/-
15 E↑X
16 RCL "SIG"
17 2
18 PI
19 ×
20 SQRT
21 ×
22 ÷
23 ×
24 END

```

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

```

Line 01: [PGM.FCN] {LBL} {NOVER} [ENTER]
Line 02: [PGM.FCN] [▲] {MVAR} {D} [ENTER]
Line 03: [PGM.FCN] [▲] {MVAR} {Q} [ENTER]
Line 04: [PGM.FCN] [▲] {MVAR} {MU} [ENTER]
Line 05: [PGM.FCN] [▲] {MVAR} {SIG} [ENTER]
Line 06: [RCL] {Q} [ENTER]
Line 07: [RCL] [-] {D} [ENTER]
Line 08: [RCL] {D} [ENTER]
Line 09: [RCL] [-] {MU} [ENTER]
Line 10: [RCL] [÷] {SIG} [ENTER]
Line 11: [X↑2]
Line 12: [2]
Line 13: [÷]
Line 14: [+/-]
Line 15: [E↑X]
Line 16: [RCL] {SIG} [ENTER]
Line 17: [2]
Line 18: [π]
Line 19: [×]
Line 20: [√x]
Line 21: [×]
Line 22: [÷]
Line 23: [×]

```

The new space already ends in .END.; finish: [PRGM]

```

01 LBL "NSHORT"
02 MVAR "D"
03 MVAR "Q"
04 MVAR "MU"
05 MVAR "SIG"
06 RCL "D"
07 RCL- "Q"
08 RCL "D"
09 RCL- "MU"
10 RCL÷ "SIG"
11 X↑2
12 2
13 ÷
14 +/-
15 E↑X
16 RCL "SIG"

```

```

17 2
18 PI
19 x
20 SQRT
21 x
22 ÷
23 x
24 END

```

Start: [GT0] [.] [.] [PRGM] (new program space, then Program-entry mode)

```

Line 01: [PGM.FCN] {LBL} {NSHORT} [ENTER]
Line 02: [PGM.FCN] [▲] {MVAR} {D} [ENTER]
Line 03: [PGM.FCN] [▲] {MVAR} {Q} [ENTER]
Line 04: [PGM.FCN] [▲] {MVAR} {MU} [ENTER]
Line 05: [PGM.FCN] [▲] {MVAR} {SIG} [ENTER]
Line 06: [RCL] {D} [ENTER]
Line 07: [RCL] [-] {Q} [ENTER]
Line 08: [RCL] {D} [ENTER]
Line 09: [RCL] [-] {MU} [ENTER]
Line 10: [RCL] [÷] {SIG} [ENTER]
Line 11: [X↑2]
Line 12: [2]
Line 13: [÷]
Line 14: [+/-]
Line 15: [E↑X]
Line 16: [RCL] {SIG} [ENTER]
Line 17: [2]
Line 18: [π]
Line 19: [×]
Line 20: [√x]
Line 21: [×]
Line 22: [÷]
Line 23: [×]

```

The new space already ends in .END.; finish: [PRGM]

Item	Procedure
parameters	MU=100, SIG=15; h=2; b=8
NOVER limits	10 to Q
NSHORT limits	Q to 190
ACC	1E-8; tighten and compare
cost	2×Xover + 8×Xshort
tail check	move truncation to $\mu \pm 7\sigma$ and confirm reported digits
candidate sensitivity	Q=90,100,110 table in §40

Interpretation. The lowest of a few candidate costs is evidence for a better candidate, not a continuous optimizer. A separate transparent optimization method would be required to claim an optimum.

49 Bonus extreme — Nonuniform spherical body

The radial-shell mass calculation in §41 shows a second astrophysical pattern: geometry contributes a $4\pi r^2$ factor, so central density alone does not dominate volume. More difficult gravitational field calculations may require separate interior and exterior shell treatments; never integrate through a geometric singularity without an analytic derivation.

PART APPENDICES

QUICK REFERENCE A-M

Operational checklists, templates, glyphs, diagnostics, validation notes and glossary.

Appendix A — One-page $\int f(x)$ checklist

30. Write the accumulated quantity and its units.
31. Choose the integration variable and finite limits.
32. Inspect domain, signs, symmetry, jumps, singularities, peaks and oscillations.
33. Estimate sign and magnitude.
34. Create/select an integrand program with global LBL and contiguous MVAR lines.
35. Store constants; press the variable soft key without a number to select it.
36. Store LLIM, ULIM and ACC; press $\int$.
37. Record X and Y.
38. Tighten ACC and/or split the interval.
39. Compare with an independent analytic, computational or dimensional check.
40. Round to meaningful digits.

Appendix B — Keystroke legend

Form	Meaning
[GTO] [.] [.] [PRGM]	new program space, then Program-entry mode
[PGM.FCN] [$\blacktriangle$] {MVAR} (X) [ENTER]	insert MVAR "X"
[$\int f(x)$]	shifted integration application key
{NAME}	press the top-row key under NAME
value {NAME}	store value into a menu variable
{NAME} with no pending number	select integration variable
[X<>Y]	exchange X and Y to inspect/restore uncertainty
1 [E] [+/-] 8	enter $1\text{E}-8$

Appendix C — ACC quick guide

Observation	Interpretation	Next move
X stable; Y shrinks	healthy convergence evidence	report only justified digits
X changes beyond Y	Y did not capture total error	split or reformulate
Y large relative to X	weak relative convergence or cancellation	integrate lobes separately
run time explodes	difficult feature or excessive ACC	inspect scale/domain; loosen ACC appropriately

Observation	Interpretation	Next move
ACC=0	near decimal128 floor in current core	use only as a deliberate experiment
Y=0	successive modeled estimates identical at working precision	not proof of exact mathematical truth

Appendix D — Reusable program templates

```

01 LBL "MODEL"
02 MVAR "X"
03 MVAR "A"
04 MVAR "B"
05 RCL "X"
06 ... calculate f(X,A,B) ...
07 END

```

Smooth finite interval: use the template directly. Piecewise model: prefer separate programs or separate interval runs before adding conditional logic. Improper model: derive and validate a finite truncation or substitution first.

Appendix E — DM42 glyph and encoder mapping

Displayed glyph / form	SwissMicros encoder input
$R \uparrow$	R^
$R \downarrow$	R\w or Rv
$\times$	\x in text; x or * in instruction mnemonics
$\div$	\:- in text; / in instruction mnemonics
$\sqrt{}$	\w/
$\int$	\S
Σ	\GS
π	\PI
$\leq$	\<=
$\geq$	\>=
$\neq$	\#
$\rightarrow$	\->
$\leftarrow$	\<-

Displayed glyph / form	SwissMicros encoder input
↓	\v
μ	\m
◦	\o
E	\E
•	\.

Primary listings in this handbook use the calculator-style Unicode glyphs. Encoder spellings are provided only for exchanging program text or checking unusual characters.

Appendix F — RPN expression patterns

Expression	Reliable stack sequence
$e^{(-x^2)}$	RCL X X↑2 +/- E↑X
$a+bx$	RCL X RCL× B RCL+ A
$1/\sqrt{1-k^2\sin^2x}$	RCL X SIN X↑2 RCL K X↑2 × 1 X<>Y – SQRT 1/X
normal density	standardize; square; ÷2; change sign; E↑X; divide by $\sigma\sqrt{2\pi}$
$x^3/(e^x-1)$	RCL X 3 Y↑X RCL X E↑X 1 – ÷

Appendix G — Post-integration stack

Register	Content
X	integral approximation
Y	current implementation's last-change estimate
Z/T	older stack contents shifted according to stack-lift mode; not a portable result interface

Appendix H — Troubleshooting

Symptom	Likely cause	Action
program absent from $\int f(x)$	no global LBL or no contiguous MVAR	inspect first lines and END
wrong trig result	DEG/RAD mismatch	set mode before integrating
Nonexistent / invalid data	missing LLIM, ULIM, variable or wrong type	re-enter real values
long run	tight ACC, cancellation, peak, tail or nonsmooth function	stop safely; inspect and split
plausible but wrong	missed feature or wrong formula	analytic limit, units and independent calculation
overflow / underflow	poor scaling or extreme exponent	normalize or change variables
sign opposite expectation	reversed limits or sign error	check ULIM–LLIM and program stack

Appendix I — Singularity decision tree

41. Is f finite and defined throughout the closed interval? If yes, continue with ordinary checks.
42. If not, identify whether the feature is removable, a jump, an integrable endpoint singularity, an internal pole or a nonintegrable endpoint.
43. For a jump, split. For a removable point, use the limiting definition or an equivalent expression.
44. For an integrable endpoint singularity, derive and test a substitution.
45. For a pole or divergent endpoint, do not report an ordinary integral. State any alternative definition explicitly.

Appendix J — Significant figures

Evidence	Digits you may report
X stable to n decimals under tighter ACC and splitting	at most those stable decimals
Y affects the n th decimal	usually stop before that uncertain digit
inputs known to p significant figures	normally no more than p , absent a full uncertainty analysis
model approximate	state model limits; extra calculator digits are cosmetic

Appendix K — Constants and conversions used

Constant	Value used	Note
c	299792458 m/s	exact SI
h	6.62607015E-34 J·s	exact SI
k ^B	1.380649E-23 J/K	exact SI
g	9.80665 m/s ²	standard gravity, exact conventional value
R	8.314462618 J/(mol·K)	example gas constant value
Hubble conversion	1/H ₀ = 977.79222168/H ₀ Gyr when H ₀ in km·s ⁻¹ ·Mpc ⁻¹	uses standard unit conversion

UNITS CHECK Exact constants do not make an approximate model exact. Measured values such as H₀, density, temperature and rate coefficients carry their own uncertainty and context.

Appendix L — Validation record

All numerical tables were generated from a machine-readable validation record saved during document construction. References use 60-decimal-digit arithmetic. Representative X/Y pairs use a direct transcription of Free42 3.3.10 core_math1.cc's Romberg loop, including its endpoint-avoiding substitution, five-estimate extrapolation window, relative stopping test and refinement cap.

Validation layer	Method
formula	independent analytic derivation or high-precision quadrature
DM42 algorithm	source-level model of current published core
program	manual RPN stack audit and analytic special cases
units	dimensional analysis in every physical example
sensitivity	ACC, split, tail or parameter sweep in delicate examples

Reproducibility caveat. The model predicts the algorithm's behavior, but a physical DM42 may differ in last displayed digits or run time because decimal128 transcendental routines and firmware can change. Verify IDENT and rerun critical cases on the target calculator.

Appendix M — Glossary

Term	Meaning
ACC	stored factor controlling the current integrator's relative convergence test
absolute error	approximation–reference
cancellation	loss of relative significance when large opposite contributions nearly cancel
convergence	tendency of refined estimates toward a stable value
improper integral	integral with infinite limit or unbounded integrand, defined through a limit
integrand	function being accumulated
quadrature	numerical evaluation of a definite integral
Romberg integration	refinement plus Richardson extrapolation
signed area	positive contribution above and negative contribution below the axis
tail	portion beyond a finite truncation
Y estimate	current INTEG last-change estimate returned in Y

References and research basis

1. SwissMicros, DM42 User Manual v3.42, 24 September 2026.
https://technical.swissmicros.com/dm42/doc/dm42_user_manual.html
2. SwissMicros, DM42 Quick Reference Guide: Numerical Integration and Variable Menus.
<https://technical.swissmicros.com/dm42/fat/HELP/dm42help.htm>
3. SwissMicros, DM42 Functions documentation.
<https://technical.swissmicros.com/dm42/fat/HELP/dm42functs.htm>
4. SwissMicros, DM42 Program Firmware History; v3.26 uses Free42 3.3.10.
<https://technical.swissmicros.com/dm42/firmware/history.html>
5. SwissMicros, DM42 Encoder/Decoder instructions and character mappings.
<https://technical.swissmicros.com/dm42/decoder/>
6. SwissMicros Free42 fork, core_math1.cc, version 3.3.10 source used to verify INTEG and ACC behavior.
<https://github.com/swissmicros/free42>
7. Hewlett-Packard, HP-42S Owner's Manual, 1988. <https://literature.hpcalc.org/community/hp42s-om-en.pdf>
8. Hewlett-Packard, HP-42S Programming Examples & Techniques, Chapter 4 Integration, 1988.
<https://literature.hpcalc.org/community/hp42s-prog-en.pdf>

9. Thomas Okken, Free42 and release history. <https://thomasokken.com/free42/>
10. NIST, Fundamental Physical Constants. <https://physics.nist.gov/cuu/Constants/>
11. NIST/SEMATECH e-Handbook of Statistical Methods, normal distribution. <https://www.itl.nist.gov/div898/handbook/>
12. Hogg, D. W., Distance measures in cosmology, arXiv:astro-ph/9905116. <https://arxiv.org/abs/astro-ph/9905116>
13. Planck's law and blackbody radiation: NIST and standard radiometry definitions. <https://physics.nist.gov/>

CAUTION This handbook is educational. Consequential engineering, financial, scientific or medical work requires current domain standards, validated data, uncertainty analysis and qualified professional judgment.